Contents

1 Basic

1.1 Default Code [d41d8c]

```cpp
#include <bits/stdc++.h>

using namespace std;
using ll = long long;

namespace rgs = ranges;

template<class T> bool
    chmin(T &a, const T &b) { return a > b && (a = b, true); }
template<class T> bool
    chmax(T &a, const T &b) { return a < b && (a = b, true); }

void solve() {

}

int main() {
  ios::sync_with_stdio(false);
  cin.tie(nullptr);

  int t = 1;
  cin >> t;
  while (t--) {
    solve();
  }

  return 0;
}
```

1.2 Debug [33ccce]

```bash
# 對拍
CODE1="a"
CODE2="ac"
set -e
g++ $CODE1.cpp -o $CODE1
g++ $CODE2.cpp -o $CODE2
g++ gen.cpp -o gen
for ((i=1;;i++))
do
  echo "--- Testing: Case #$i ---"
  ./gen > input
  # python3 gen.py > input
  ./$CODE1 < input > $CODE1.out
  ./$CODE2 < input > $CODE2.out
  cmp $CODE1.out $CODE2.out || break
done
# 多重解, ifstream in(argv[1]);
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ gen.cpp -o gen
g++ checker.cpp -o checker
for ((i=1;;i++))
do
  ./gen > input
  ./$CODE < input > $CODE.out
  ./checker $CODE.out < input || break
# 互動
CODE="a"
set -e
g++ $CODE.cpp -o $CODE
g++ checker.cpp -o checker
PIPE_IN="in"
PIPE_OUT="out"
trap 'rm -f $PIPE_IN $PIPE_OUT' EXIT
mkfifo $PIPE_IN $PIPE_OUT
for ((i=1;;i++))
do
  echo "--- Testing: Case #$i ---"
  ./$CODE < $PIPE_IN > $PIPE_OUT &
  (exec 3>$PIPE_IN 4<$PIPE_OUT; ./checker <&4 >&3) || break
done
# 參考 checker
ll AC(int n) { return ans; }
void WA(string log = "") { cerr << log << endl; exit(1); }
void checkAC(string log = "") {
  string trash;
  if (cin >> trash) WA("redundant output\n" + log);
}
int main() {
  int n = uniform_int_distribution<int>(1, 10)(rng);
  ll sol = AC(n);
  stringstream log;
  cout << n << endl;
  log << n << endl;
  log << "judge: " << endl;
  log << "team: " << endl;
  WA(log.str());
  checkAC();
  return 0;
}
```

1.3 Compare Fuction [d41d8c]

```cpp
// 1. sort, 二分搜刻在函式內 lambda 就好
// 2. priority queue 小到大是 >, set 是 <
// 3. set 不能 = , multiset 必須 =
// 4. 確保每個成員都要比到
// 5. pbds_multiset 不要用 lower_bound
// 6. 如果要用 find, 插入 inf 後使用 upper_bound
// 7. multiset 可以跟 set 一樣使用, 但請注意第 3、4 點
auto cmp = [](int i, int j) { return i > j; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> a {1, 2, 5, 4, 3}; // 小心不要改到 a
auto cmp = [&a](int i, int j) { return a[i] > a[j]; };
priority_queue<int, vector<int>, decltype(cmp)> pq(cmp);

vector<int> v {1, 2, 3, 4, 5};
upper_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first b that a < b, a is 2
lower_bound(v.begin(), v.end(), 2, [](int a, int b)
{ return a < b; }); // find first a that a < b fail, b is 2
```

1.4 Pbds [d41d8c]

```cpp
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template<class T, class F = less<T>>
using pbds_set = tree<T, null_type
    , F, rb_tree_tag, tree_order_statistics_node_update>;
template<class T, class F = less_equal<T>>
using pbds_multiset = tree<T, null_type
    , F, rb_tree_tag, tree_order_statistics_node_update>;
```

1.5 Int128 [eb8585]

```
using i128 = __int128_t; // 1.7E38
istream &operator>>(istream &is, i128 &a) {
  i128 sgn = 1; a = 0;
  string s; is >> s;
  for (auto c : s) {
    if (c == '-') sgn = -1;
    else a = a * 10 + c - '0';
  }
  a *= sgn;
  return is;
}
ostream &operator<<(ostream &os, i128 a) {
  if (a == 0) return os << 0;
  string res;
  if (a < 0) os << '-', a = -a;
  while (a) res.push_back(a % 10 + '0'), a /= 10;
  reverse(res.begin(), res.end());
  os << res;
  return os;
}
```

2 Graph

2.1 Binary Lifting LCA [4399db]

```
const int N = 2E5;
const int Lg = __lg(N); // __lg(max(N, Qi)), [0, Lg]
int up[N][Lg + 1];
vector<int> dep, dfn;
void build(int n, vector<vector<int>> &g, int rt = 0) {
  dep.assign(n, 0); dfn.assign(n, 0);
  int cur = 0;
  [&](this auto &&self, int x, int p) -> void {
    dfn[x] = cur++;
    up[x][0] = p;
    for (int i = 1; i <= Lg; i++) {
      int nxt = up[x][i - 1];
      up[x][i] = up[nxt][i - 1];
    }
    for (auto y : g[x]) {
      if (y == p) continue;
      up[y][0] = x;
      dep[y] = dep[x] + 1;
      self(y, x);
    }
  } (rt, rt);
}
int lca(int a, int b) {
  if (dep[a] < dep[b]) swap(a, b);
  int pull = dep[a] - dep[b];
  for (int i = 0; i <= Lg; i++)
    if (pull >> i & 1) a = up[a][i];
  if (a == b) return a;
  for (int i = Lg; i >= 0; i--)
    if (up[a][i] != up[b][i])
      a = up[a][i], b = up[b][i];
  return up[a][0];
}
int jump(int x, int k) {
  for (int i = Lg; i >= 0; i--)
    if (k >> i & 1) x = up[x][i];
  return x;
}
int dist(int a, int b) {
  return dep[a] + dep[b] - 2 * dep[lca(a, b)];
}
```

2.2 Heavy Light Decomposition [dc98bc]

```
struct HLD {
  int n, cur = 0;
  vector<int> siz, top, dep, par, in, out, seq;
  vector<vector<int>> adj;
  HLD(int n) : n(n) {
    siz.resize(n); top.resize(n); dep.resize(n);
    par.resize(n); in.resize(n); out.resize(n);
    seq.resize(n); adj.assign(n, {});
  }
  void addEdge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
  }
  void work(int rt = 0) {
    top[rt] = rt;
    dep[rt] = 0;
    par[rt] = -1;
    dfs1(rt); dfs2(rt);
  }
  void dfs1(int u) {
    if (par[u] != -1) adj
        [u].erase(find(adj[u].begin(), adj[u].end(), par[u]));
    siz[u] = 1;
    for (auto &v : adj[u]) {
      par[v] = u, dep[v] = dep[u] + 1;
      dfs1(v);
      siz[u] += siz[v];
      if (siz[v] > siz[adj[u][0]]) {
        swap(v, adj[u][0]);
      } // 讓 adj[u][0] 是重子節點
    }
  }
  void dfs2(int u) {
    in[u] = cur++;
    seq[in[u]] = u; // dfn 對應的編號
    for (auto v : adj[u]) {
      top[v] = v == adj[u][0] ? top[u] : v;
      dfs2(v);
    }
    out[u] = cur;
  }
  int lca(int u, int v) {
    while (top[u] != top[v]) {
      if (dep[top[u]] > dep[top[v]]) {
        u = par[top[u]];
      } else {
        v = par[top[v]];
      }
    }
    return dep[u] < dep[v] ? u : v;
  }
  int dist(int u, int v) {
    return dep[u] + dep[v] - 2 * dep[lca(u, v)];
  }
  int jump(int u, int k) {
    if (dep[u] < k) return -1;
    int d = dep[u] - k;
    while (dep[top[u]] > d) u = par[top[u]];
    return seq[in[u] - dep[u] + d];
  }
  bool isAncester(int u, int v) {
    return in[u] <= in[v] && in[v] < out[u];
  }
  int rootedParent(int rt, int v) {
    if (rt == v) return rt;
    if (!isAncester(v, rt)) return par[v];
    auto it = upper_bound(adj[v].begin(), adj[v].end(), rt,
      [&](int x, int y) {
        return in[x] < in[y];
      }) - 1;
    return *it;
  }
  int rootedSize(int rt, int v) {
    if (rt == v) return n;
    if (!isAncester(v, rt)) return siz[v];
    return n - siz[rootedParent(rt, v)];
  }
  int rootedLca(int rt, int a, int b) {
    return lca(rt, a) ^ lca(a, b) ^ lca(b, rt);
  }
};
```

2.3 Centroid Decomposition [7ef37f]

```
vector<bool> vis(n);
vector<int> siz(n), par(n, -1);
auto findSize = [&](auto &&self, int u, int p) -> int {
  siz[u] = 1;
  for (int v : g[u]) {
    if (v == p || vis[v]) continue;
    siz[u] += self(self, v, u);
  }
  return siz[u];
};
auto findCen = [&](auto &&self, int u, int p, int sz) -> int {
  for (int v : g[u]) {
    if (v == p || vis[v]) continue;
    if (siz[v] * 2 > sz) return self(self, v, u, sz);
  }
  return u;
};
auto build = [&](auto &&self, int u, int p) -> void {
  findSize(findSize, u, p);
  int c = findCen(findCen, u, -1, siz[u]);
  vis[c] = true, par[c] = p;
  for (int v : g[c]) if (!vis[v]) self(self, v, c);
}; build(build, 0, -1);
```

2.4 SCC [fae635]

```
struct SCC {
  int n, cur = 0, cnt = 0;
  vector<vector<int>> adj;
  vector<int> stk, dfn, low, bel;
  SCC(int n) : n(n), adj(n), dfn(n, -1), low(n), bel(n, -1) {}
  void addEdge(int u, int v) { adj[u].push_back(v); }
  void dfs(int x) {
    dfn[x] = low[x] = cur++;
    stk.push_back(x);
    for (auto y : adj[x]) {
      if (dfn[y] == -1) {
        dfs(y);
        low[x] = min(low[x], low[y]);
      } else if (bel[y] == -1) {
        low[x] = min(low[x], dfn[y]);
      }
    }
    if (dfn[x] == low[x]) {
      int y;
      do {
        y = stk.back();
        bel[y] = cnt;
```

```
        stk.pop_back();
      } while (y != x);
      cnt++;
    }
  }
  vector<int> work() {
    for (int i = 0; i < n; i++)
      if (dfn[i] == -1) dfs(i);
    return bel;
  }
  struct Graph {
    int n;
    vector<pair<int, int>> edges;
    vector<int> siz, cnte;
  };
  Graph compress() {
    Graph g; g.n = cnt;
    g.siz.resize(cnt); g.cnte.resize(cnt);
    for (int i = 0; i < n; i++) {
      g.siz[bel[i]]++;
      for (auto j : adj[i]) {
        if (bel[i] != bel[j]) {
          g.edges.emplace_back(bel[i], bel[j]);
        } else {
          g.cnte[bel[i]]++;
        }
      }
    }
    return g;
  }
};
```

2.5 2-SAT [093413]

```
struct TwoSat : SCC {
  vector<int> ans;
  TwoSat(int n) : SCC(2 * n), ans(n) {}
  void addClause(int u, bool a, int v, bool b) {
    addEdge(2 * u + !a, 2 * v + b);
    addEdge(2 * v + !b, 2 * u + a);
  }
  void ifThen(int u, bool a, int v, bool b) {
    // 必取 A: not A -> A
    addEdge(2 * u + a, 2 * v + b);
  }
  bool work() {
    SCC::work(); // bel[!a] > bel[a] => !a -> a
    for (int i = 0; i < n / 2; i++) {
      if (bel[2 * i] == bel[2 * i + 1]) return false;
      ans[i] = bel[2 * i] > bel[2 * i + 1];
    }
    return true;
  }
};
```

2.6 EBCC [a84a20]

```
struct EBCC { // CF/contest/1986/pF
  int n, cur = 0, cnt = 0;
  vector<vector<int>> adj;
  vector<int> stk, dfn, low, bel;
  vector<pair<int, int>> bridges; // 關鍵邊
  EBCC(int n) : n(n), adj(n), low(n), dfn(n, -1), bel(n, -1) {}
  void addEdge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
  }
  void dfs(int x, int p) {
    dfn[x] = low[x] = cur++;
    stk.push_back(x);
    for (auto y : adj[x]) {
      if (y == p) continue;
      if (dfn[y] == -1) {
        dfs(y, x);
        low[x] = min(low[x], low[y]);
        if (low[y] > dfn[x]) {
          bridges.emplace_back(x, y);
        }
      } else if (bel[y] == -1) {
        low[x] = min(low[x], dfn[y]);
      }
    }
    if (dfn[x] == low[x]) {
      int y;
      do {
        y = stk.back();
        bel[y] = cnt;
        stk.pop_back();
      } while (y != x);
      cnt++;
    }
  }
  vector<int> work() { // not connected
    for (int i = 0; i < n; i++)
      if (dfn[i] == -1) dfs(i, -1);
    return bel;
  }
  struct Graph {
    int n;
    vector<pair<int, int>> edges;
    vector<int> siz, cnte;
  };
  Graph compress() {
    Graph g; g.n = cnt;
    g.siz.resize(cnt); g.cnte.resize(cnt);
    for (int i = 0; i < n; i++) {
      g.siz[bel[i]]++;
      for (auto j : adj[i]) {
        if (bel[i] < bel[j]) {
          g.edges.emplace_back(bel[i], bel[j]);
        } else if (bel[i] == bel[j] && i < j) {
          g.cnte[bel[i]]++;
        }
      }
    }
    return g;
  }
};
```

2.7 VBCC [775dfa]

```
struct VBCC {
    int n, cur = 0, cnt = 0;
    vector<vector<int>> adj, bcc;
    vector<int> dfn, low, stk;
    vector<bool> ap;
    VBCC(int n) : n(n), adj(n), dfn(n, 0), low(n, 0), ap(n) {}
    void addEdge(int u, int v) {
        adj[u].push_back(v);
        adj[v].push_back(u);
    }
    void dfs(int x, int p = -1) {
        int ch = 0;
        dfn[x] = low[x] = ++cur;
        stk.push_back(x);
        for (int y : adj[x]) {
            if (!dfn[y]) {
                dfs(y, x);
                ch++;
                low[x] = min(low[x], low[y]);
                if (dfn[x] <= low[y]) {
                    ap[x] = true;
                    bcc.emplace_back();
                    int v;
                    do {
                        v = stk.back();
                        stk.pop_back();
                        bcc.back().push_back(v);
                    } while (v != y);
                    bcc.back().push_back(x);
                    cnt++;
                }
            } else if (dfn[y] < dfn[x] && y != p)
                low[x] = min(low[x], dfn[y]);
        }
        if (p == -1 && ch < 2) ap[x] = false;
    }
    void work() {
        for (int i = 0; i < n; i++)
            if (!dfn[i])
                if (adj[i].empty()) {
                    bcc.emplace_back(1, i);
                    cnt++;
                } else dfs(i);
    }
    struct Graph {
        int n;
        vector<pair<int, int>> edges;
        vector<int> bel, siz, cnte;
    };
    Graph compress() {
        Graph g;
        g.bel.assign(n, -1);
        g.siz.assign(cnt, 0);
        g.cnte.assign(cnt, 0);
        g.n = cnt;
        for (int i = 0; i < n; i++) {
            if (ap[i]) {
                g.bel[i] = g.n++;
                g.siz.push_back(1);
                g.cnte.push_back(0);
            }
        }
        vector<bool> in_bcc(n);
        for (int i = 0; i < cnt; i++) {
            for (int u : bcc[i]) in_bcc[u] = true;
            int edges = 0;
            for (int u : bcc[i]) {
                if (ap[u]) g.edges.emplace_back(g.bel[u], i);
                else g.bel[u] = i, g.siz[i]++;
                for (int v : adj[u]) if (in_bcc[v]) edges++;
            }
            g.cnte[i] = edges / 2;
            for (int u : bcc[i]) in_bcc[u] = false;
        }
        return g;
    }
};
```

2.8 Functional Graph [060d10]

```
const int N = 2E5;
const int Lg = __lg(N); // __lg(max(n, qi)), [0, Lg]
int cht[N][Lg];
```

```cpp
struct FuntionalGraph {
  int n, cnt = 0;
  vector<int> g, bel, id, cycsz, in, top, hei;
  FuntionalGraph(const vector<int> &g) : n(g.size()
      ), g(g), bel(n, -1), id(n), in(n), top(n, -1), hei(n) {
    for (int i = 0; i < n; i++)
      cht[i][0] = g[i], in[g[i]]++;
    for (int i = 1; i <= Lg; i++)
      for (int u = 0; u < n; u++) {
        int nxt = cht[u][i - 1];
        cht[u][i] = cht[nxt][i - 1];
      }
    for (int i = 0; i < n; i++)
      if (in[i] == 0) label(i);
    for (int i = 0; i < n; i++)
      if (top[i] == -1) label(i);
  }
  void label(int u) {
    vector<int> p; int cur = u;
    while (top[cur] == -1) {
      top[cur] = u;
      p.push_back(cur);
      cur = g[cur];
    }
    auto s = find(p.begin(), p.end(), cur);
    vector<int> cyc(s, p.end());
    p.erase(s, p.end()); p.push_back(cur);
    for (int i = 0; i < (int)cyc.size(); i++)
      bel[cyc[
          i]] = cnt, id[cyc[i]] = i, hei[cyc[i]] = cyc.size();
    if (!cyc.empty())
      ++cnt, cycsz.push_back(cyc.size());
    for (int i = p.size() - 1; i > 0; i--)
      id[p[i
          - 1]] = id[p[i]] - 1, hei[p[i - 1]] = hei[p[i]] + 1;
  }
  int jump(int u, int k) {
    for (int b = 0; k > 0; b++) {
      if (k & 1) u = cht[u][b];
      k >>= 1;
    }
    return u;
  }
};
```

2.9 Bellman-Ford [430de2]

```cpp
// 用 Bellman Ford 找負環
void bellmanFord() {
  int n, m; cin >> n >> m;
  vector<array<int, 3>> e;
  for (int i = 0; i < m; i++) {
    int u, v, w; cin >> u >> v >> w;
    u--, v--; e.push_back({u, v, w});
  }
  vector<ll> dis(n, inf), par(n);
  int t = -1; dis[0] = 0;
  for (int i = 1; i <= n; i++) {
    for (auto [u, v, w] : e) {
      if (dis[v] > dis[u] + w) {
        dis[v] = dis[u] + w;
        par[v] = u;
        if (i == n) t = v;
      }
    }
  }
  if (t == -1) { cout << "NO\n"; return; }
  for (int i = 1; i < n; i++) t = par[t];
  vector<int> ans {t};
  int i = t;
  do {
    i = par[i];
    ans.push_back(i);
  } while (i != t);
  reverse(ans.begin(), ans.end());
  cout << "YES\n";
  for (auto x : ans) cout << x + 1 << " ";
}
```

2.10 Euler [13ce48]

```cpp
// 1. 無向圖是歐拉圖：
// 非零度頂點是連通的
// 頂點的度數都是偶數

// 2. 無向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是連通的
// 恰有 2 個奇度頂點

// 3. 有向圖是歐拉圖：
// 非零度頂點是強連通的
// 每個頂點的入度和出度相等

// 4. 有向圖是半歐拉圖(有路沒有環)：
// 非零度頂點是弱連通的
// 至多一個頂點的出度與入度之差為 1
// 至多一個頂點的入度與出度之差為 1
// 其他頂點的入度和出度相等
vector<int> ans;
[&](this auto &&self, int u) -> void {
  while (g[u].size()) {
    int v = *g[u].begin();
    g[u].erase(v);
    self(v);
  }
  ans.push_back(u);
} (0); reverse(ans.begin(), ans.end());
```

2.11 Virtual Tree [b2b0eb]

```cpp
// 多次詢問給某些關鍵點，虛樹可達成快速樹 DP (前處理每個點)
// 例如這題是有權樹，給一些關鍵點，求跟 vertex 1 隔開的最小成本
// 前處理 root 到所有點的最小邊權
void solve(int n, vector<vector<pair<int, int>>> &g, int q) {
  build(n, g); // build LCA
  vector<int> dis(n, 1E9); // root 到各點的最小邊權
  [&](this auto &&self, int x, int p) -> void {
    for (auto [y, w] : g[x]) {
      if (y == p) continue;
      dis[y] = min(w, dis[x]);
      self(y, x);
    }
  } (0, -1);
  vector<vector<int>> vt(n);
  vector<int> isKey(n);
  vector<ll> dp(n);
  while (q--) {
    int m; cin >> m;
    vector<int> key(m);
    for (int i = 0; i < m; i++) {
      cin >> key[i];
      isKey[--key[i]] = 1;
    }
    key.push_back(0); // 固定 0 為 root，看題目需求
    sort(key.begin(), key.end(), [&](int a, int b) {
      return dfn[a] < dfn[b];
    }); // 要 sort 再 insert
    vector<int> stk; // 維護一條虛樹鏈
    for (int k : key) {
      if (stk.empty()) {
        stk.push_back(k);
        continue;
      }
      int l = lca(stk.back(), k);
      if (l == stk.back()) { // 直接一條鏈
        stk.push_back(k);
        continue;
      } // 次頂是 lca 的子孫，直接跟最頂連邊
      while (stk
          .size() > 1 && dfn[stk[stk.size() - 2]] > dfn[l]) {
        vt[stk[stk.size() - 2]].push_back(stk.back());
        stk.pop_back();
      } // 保證 back 是 lca 的子孫，且次頂不是 lca 的子孫
      vt[l].push_back(stk.back());
      if (stk.size
          () < 2 || stk[stk.size() - 2] != l) stk.back() = l;
      else stk
          .pop_back(); // lca 的子孫已經用不到了，直接換成 lca
      stk.push_back(k);
    }
    while (stk.size() > 1) {
      vt[stk[stk.size() - 2]].push_back(stk.back());
      stk.pop_back();
    }
    int rt = stk[0]; stk.clear();
    [&](this auto &&self, int x) -> void {
      for (auto y : vt[x]) {
        self(y);
        if (isKey[y]) { // 直接砍了
          dp[x] += dis[y];
        } else { // 不砍 or 砍
          dp[x] += min<ll>(dp[y], dis[y]);
        } // 記得 reset
        isKey[y] = dp[y] = 0;
      }
      vt[x].clear(); // 記得 reset
    } (rt);
    cout << dp[rt] << "\n";
    isKey[rt] = 0;
    dp[rt] = 0; // 最後 reset root
  }
}
```

2.12 Dominator Tree [820fdd]

```cpp
// dom
    存起點到達此點的必經的上個節點(起點 = 自己), 無法到達 = -1
struct DominatorTree {
  int n, id;
  vector<vector<int>> adj, radj, bucket;
  vector<int> sdom, dom, vis, rev, pa, rt, mn, res;
  DominatorTree(int n) : n(n) {
    sdom.resize(n), rev.resize(n);
    pa.resize(n), rt.resize(n);
    mn.resize(n), res.resize(n);
    bucket.assign(n, {});
    adj.assign(n, {}), radj.assign(n, {});
    dom.assign(n, -1), vis.assign(n, -1);
```

```cpp
  }
  void addEdge(int u, int v) { adj[u].push_back(v); }
  int query(int v, int x) {
    if (rt[v] == v) return x ? -1 : v;
    int p = query(rt[v], 1);
    if (p == -1) return x ? rt[v] : mn[v];
    if (sdom[mn[v]] > sdom[mn[rt[v]]])
      mn[v] = mn[rt[v]];
    rt[v] = p;
    return x ? p : mn[v];
  }
  void dfs(int v) {
    vis[v] = id, rev[id] = v;
    rt[id] = mn[id] = sdom[id] = id, id++;
    for (int u : adj[v]) {
      if (vis[u] == -1)
        dfs(u), pa[vis[u]] = vis[v];
      radj[vis[u]].push_back(vis[v]);
    }
  }
  vector<int> build(int s) {
    dfs(s);
    for (int i = id - 1; i >= 0; i--) {
      for (int u : radj[i])
        sdom[i] = min(sdom[i], sdom[query(u, 0)]);
      if (i) bucket[sdom[i]].push_back(i);
      for (int u : bucket[i]) {
        int p = query(u, 0);
        dom[u] = sdom[p] == i ? i : p;
      }
      if (i) rt[i] = pa[i];
    }
    res.assign(n, -1);
    for (int i = 1; i < id; i++)
      if (dom[i] != sdom[i])
        dom[i] = dom[dom[i]];
    for (int i = 1; i < id; i++)
      res[rev[i]] = rev[dom[i]];
    res[s] = s;
    for (int i = 0; i < n; i++)
      dom[i] = res[i];
    return dom;
  }
};
```

3 Data Structure

3.1 DSU [d41d8c]

```cpp
struct DSU {
  int n;
  vector<int> f, siz;
  DSU(int n) : n(n), f(n), siz(n, 1)
  { iota(f.begin(), f.end(), 0); }
  int find(int x) { return x == f[x] ? x : f[x] = find(f[x]); }
  bool same(int x, int y) { return find(x) == find(y); }
  bool merge(int x, int y) {
    x = find(x), y = find(y);
    if (x == y) return false;
    if (siz[x] < siz[y]) swap(x, y);
    siz[x] += siz[y];
    f[y] = x;
    n--;
    return true;
  }
  int size(int x) { return siz[find(x)]; }
};
struct DSU {
  int n;
  vector<int> f, siz, stk;
  DSU(int n) : n(n), f(n), siz(n, 1)
  { iota(f.begin(), f.end(), 0); }
  int find(int x) { return x == f[x] ? x : find(f[x]); }
  bool same(int x, int y) { return find(x) == find(y); }
  bool merge(int x, int y) {
    x = find(x), y = find(y);
    if (x == y) return false;
    if (siz[x] < siz[y]) swap(x, y);
    siz[x] += siz[y];
    f[y] = x;
    n--;
    stk.push_back(y);
    return true;
  }
  void undo(int x) {
    while (stk.size() > x) {
      int y = stk.back();
      stk.pop_back();
      n++;
      siz[f[y]] -= siz[y];
      f[y] = y;
    }
  }
  int size(int x) { return siz[find(x)]; }
};
```

3.2 Fenwick [d41d8c]

```cpp
template<class T> struct Fenwick {
  int n; vector<T> a;
  Fenwick(int n) : n(n), a(n) {}
  void add(int x, const T &v) {
    for (int i = x + 1; i <= n; i += i & -i)
      a[i - 1] = a[i - 1] + v;
  }
  T sum(int x) {
    T ans{};
    for (int i = x; i > 0; i -= i & -i)
      ans = ans + a[i - 1];
    return ans;
  }
  T rangeSum(int l, int r) { return sum(r) - sum(l); }
  int select(const T &k, int start = 0) {
    // 找到最小的 x, 使得 sum(x + 1) - sum(start) > k
    // prefix sum 要有單調性
    int x = 0; T cur = -sum(start);
    for (int i = 1 << __lg(n); i; i /= 2)
      if (x + i <= n && cur + a[x + i - 1] <= k)
        x += i, cur = cur + a[x - 1];
    return x;
  }
};
template<class T> struct Fenwick2D {
  int n, m;
  vector<vector<T>> a;
  Fenwick2D(int n, int m) : n(n), m(m), a(n, vector<T>(m)) {}
  void add(int x, int y, const T &v) {
    for (int i = x + 1; i <= n; i += i & -i)
      for (int j = y + 1; j <= m; j += j & -j)
        a[i - 1][j - 1] = a[i - 1][j - 1] + v;
  }
  T sum(int x, int y) {
    T ans{};
    for (int i = x; i > 0; i -= i & -i)
      for (int j = y; j > 0; j -= j & -j)
        ans = ans + a[i - 1][j - 1];
    return ans;
  }
  T rangeSum(int x1, int y1, int x2, int y2)
  { return
      sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1); }
};
```

3.3 Range Fenwick [d41d8c]

```cpp
template<class T> struct RangeFenwick {
  int n;
  vector<T> d, di;
  RangeFenwick(int n) : n(n), d(n), di(n) {}
  void add(int x, const T &v) {
    T vi = v * (x + 1);
    for (int i = x + 1; i <= n; i += i & -i)
      d[i - 1] = d[i - 1] + v, di[i - 1] = di[i - 1] + vi;
  }
  void rangeAdd
      (int l, int r, const T &v) { add(l, v), add(r, -v); }
  T sum(int x) {
    T ans{};
    for (int i = x; i > 0; i -= i & -i)
      ans = ans + T(x + 1) * d[i - 1] - di[i - 1];
    return ans;
  }
  T rangeSum(int l, int r) { return sum(r) - sum(l); }
  int select(const T &k, int start = 0) {
    int x = 0;
    T cur {}, curi {}, sub = sum(start);
    for (int i = 1 << __lg(n); i; i /= 2) {
      if (x + i <= n && T(x + i + 1) * (cur +
          d[x + i - 1]) - (curi + di[x + i - 1]) - sub <= k) {
        x += i;
        cur = cur + d[x - 1], curi = curi + di[x - 1];
      }
    }
    return x;
  }
};
template<class T> struct RangeFenwick2D {
  int n, m;
  vector<vector<T>> d, di, dj, dij;
  RangeFenwick2D
      (int n, int m) : n(n), m(m), dij(n, vector<T>(m))
  { d = di = dj = dij; }
  void add(int x, int y, const T &v) {
    T vi = v * (x + 1), vj = v * (y + 1);
    T vij = v * (x + 1) * (y + 1);
    for (int i = x + 1; i <= n; i += i & -i)
      for (int j = y + 1; j <= m; j += j & -j) {
        d[i - 1][j - 1] = d[i - 1][j - 1] + v;
        di[i - 1][j - 1] = di[i - 1][j - 1] + vi;
        dj[i - 1][j - 1] = dj[i - 1][j - 1] + vj;
        dij[i - 1][j - 1] = dij[i - 1][j - 1] + vij;
      }
  }
  void rangeAdd(int x1, int y1, int x2, int y2, const T &v) {
    add(x1, y1, v), add(x2, y2, v);
    add(x1, y2, -v), add(x2, y1, -v);
  }
  T sum(int x, int y) {
    T ans{};
    for (int i = x; i > 0; i -= i & -i)
      for (int j = y; j > 0; j -= j & -j)
        ans = ans + (x + 1) * (y + 1) * d[i - 1][j - 1]
        - (y + 1) * di[i - 1][j - 1]
        - (x + 1) * dj[i - 1][j - 1]
```

```
      + dij[i - 1][j - 1];
    return ans;
  }
  T rangeSum(int x1, int y1, int x2, int y2)
  { return
      sum(x2, y2) - sum(x1, y2) - sum(x2, y1) + sum(x1, y1); }
};
```

3.4 Lazy Segment Tree [d41d8c]

```
template<class Info, class Tag> struct LazySegmentTree {
  int n;
  vector<Info> info;
  vector<Tag> tag;
  LazySegmentTree
      (int n) : n(n), info(4 << __lg(n)), tag(4 << __lg(n)) {}
  LazySegmentTree(const vector<Info> &v
      ) : n(v.size()), info(4 << __lg(n)), tag(4 << __lg(n)) {
    auto build = [&](auto &&self, int l, int r, int p) {
      if (r - l == 1) {
        info[p] = v[l];
        return;
      }
      int m = (l + r) / 2;
      self(self, l, m, 2 * p);
      self(self, m, r, 2 * p + 1);
      pull(p);
    };
    build(build, 0, n, 1);
  }
  void pull(int p) { info[p] = info[2 * p] + info[2 * p + 1]; }
  void apply(const Tag &t, int l, int r, int p) {
    info[p].apply(t, l, r);
    tag[p].apply(t);
  }
  void push(int l, int r, int p) {
    int m = (l + r) / 2;
    apply(tag[p], l, m, 2 * p);
    apply(tag[p].
        offset(m - l), m, r, 2 * p + 1); // tag 可位移 (m - l)
    tag[p] = Tag();
  }
  void modify(int x, const Info &i) { modify(x, i, 0, n, 1); }
  void modify(int x, const Info &i, int l, int r, int p) {
    if (r - l == 1) {
      info[p] = i;
      return;
    }
    int m = (l + r) / 2;
    push(l, r, p);
    if (x < m) modify(x, i, l, m, 2 * p);
    else modify(x, i, m, r, 2 * p + 1);
    pull(p);
  }
  Info query(int ql, int qr) { return query(ql, qr, 0, n, 1); }
  Info query(int ql, int qr, int l, int r, int p) {
    if (qr <= l || ql >= r) return Info();
    if (ql <= l && r <= qr) return info[p];
    int m = (l + r) / 2;
    push(l, r, p);
    return query(
        ql, qr, l, m, 2 * p) + query(ql, qr, m, r, 2 * p + 1);
  }
  void rangeApply(int ql, int
      qr, const Tag &t) { rangeApply(ql, qr, t, 0, n, 1); }
  void rangeApply
      (int ql, int qr, const Tag &t, int l, int r, int p) {
    if (qr <= l || ql >= r) return;
    if (ql <= l && r <= qr) {
      apply(t.offset(l - ql), l, r, p); // tag 可位移 (l - ql)
      return;
    }
    int m = (l + r) / 2;
    push(l, r, p);
    rangeApply(ql, qr, t, l, m, 2 * p);
    rangeApply(ql, qr, t, m, r, 2 * p + 1);
    pull(p);
  }
  template<class F> int findFirst(int ql, int qr, F &&f) {
    return findFirst(ql, qr, f, 0, n, 1);
  } // 若要找 last，先右子樹遞迴即可
  template<class F> int
      findFirst(int ql, int qr, F &&f, int l, int r, int p) {
    if (qr <= l || ql >= r) return -1;
    if (ql <= l && r <= qr && !f(info[p])) return -1;
    if (r - l == 1) return l;
    int m = (l + r) / 2;
    push(l, r, p);
    int res = findFirst(ql, qr, f, l, m, 2 * p);
    if (res == -1) res = findFirst(ql, qr, f, m, r, 2 * p + 1);
    return res;
  }
};
struct Tag { // 有些 Tag 不用 push 例如 sweepLine
  void apply(const Tag &t) & {}
  Tag offset(int d) const { return *this; }
};
struct Info {
  void apply(const Tag &t, int l, int r) & {} /*
  Info &operator=(const Info &i) & {
    // do something... 部分 assignment 使用
    return *this;
  } */
};
Info operator+(const Info &a, const Info &b) {
  Info c;
  return c;
}
```

3.5 Persistent Segment Tree [d41d8c]

```
template<class Info> struct PST {
  struct Node {
    Info info = Info();
    int lc = 0, rc = 0;
  };
  int n;
  vector<Node> nd;
  vector<int> rt;
  PST(int n) : n(n), nd(1), rt(1) {}
  PST(const vector<Info> &v) : n(v.size()), nd(1) {
    auto build = [&](auto &&self, int l, int r) -> int {
      int p = generate();
      if (r - l == 1) {
        nd[p].info = v[l];
        return p;
      }
      int m = (l + r) / 2;
      int lc = self(self, l, m);
      int rc = self(self, m, r);
      nd[p].lc = lc;
      nd[p].rc = rc;
      pull(nd[p]);
      return p;
    };
    rt.push_back(build(build, 0, n));
  }
  void pull
      (Node &p) { p.info = nd[p.lc].info + nd[p.rc].info; }
  int copy(int p) { // copy 一個 node
    nd.push_back(nd[p]);
    return nd.size() - 1;
  }
  int generate() { // 創立新的 node
    nd.emplace_back();
    return nd.size() - 1;
  }
  void modify(int x, const Info &i, int ver = 0) {
    if (int(rt.size()) <= ver) rt.resize(ver + 1);
    rt[ver] = modify(x, i, 0, n, rt[ver]);
  }
  int modify(int x, const Info &i, int l, int r, int p) {
    p = p ? copy(p) : generate();
    if (r - l == 1) {
      nd[p].info = i;
      return p;
    }
    int m = (l + r) / 2;
    if (x < m) {
      int y = modify(x, i, l, m, nd[p].lc);
      nd[p].lc = y;
    } else {
      int y = modify(x, i, m, r, nd[p].rc);
      nd[p].rc = y;
    }
    pull(nd[p]);
    return p;
  }
  Info query(int ql, int qr, int ver = 0)
  { return query(ql, qr, 0, n, rt[ver]); }
  Info query(int ql, int qr, int l, int r, int p) {
    if (l >= qr || r <= ql || p == 0) return Info();
    if (ql <= l && r <= qr) return nd[p].info;
    int m = (l + r) / 2;
    return query(ql
        , qr, l, m, nd[p].lc) + query(ql, qr, m, r, nd[p].rc);
  }
  void createVersion(int ver) { rt.push_back(rt[ver]); }
};
```

3.6 Kth-element [d41d8c]

```
template<class T> struct StaticKth : PST<int> {
  int dct(T x) {
      return lower_bound(s.begin(), s.end(), x) - s.begin(); }
  vector<T> a, s; // array, sorted
  StaticKth(const vector<T> &a
      , const vector<T> &s) : a(a), s(s), PST<int>(s.size()) {
    assert(is_sorted(s.begin(), s.end()));
    for (int i = 0; i < a.size(); i++) {
      createVersion(i);
      add(dct(a[i]), 1, i + 1);
    }
  }
  int work(int a, int b, int l, int r, int k) {
    if (r - l == 1) return l;
    int x = nd[nd[b].lc].info - nd[nd[a].lc].info;
    int m = (l + r) / 2;
    if (x >= k) return work(nd[a].lc, nd[b].lc, l, m, k);
    else return work(nd[a].rc, nd[b].rc, m, r, k - x);
  }
  int work(int l, int r, int k) { // [l, r), k > 0
    return s[work(rt[l], rt[r], 0, n, k)];
```

```
  }
}; // Dynamic -> Fenwick 包動態開點值域線段樹
```

3.7 Link Cut Tree [d41d8c]

```
// 有用到 pathApply 才需要 apply 有關的
const int Mod = 51061;
struct Tag {
  ll add = 0, mul = 1;
  void apply(const Tag &t) {
    mul = mul * t.mul % Mod;
    add = (add * t.mul % Mod + t.add) % Mod;
  }
};
struct Info {
  int siz = 0; // 實鏈的長度
  ll val = 0, sum = 0;
  void apply(const Tag &t) {
    val = (val * t.mul % Mod + t.add) % Mod;
    sum = (sum * t.mul % Mod + t.add * siz % Mod) % Mod;
  }
  void pull(const Info &l, const Info &r) {
    siz = 1 + l.siz + r.siz;
    sum = (l.sum + r.sum + val) % Mod;
  }
};
struct LinkCutTree { // 1-based
  vector<Info> info;
  vector<Tag> tag;
  vector<array<int, 2>> ch;
  vector<int> p, rev;
  LinkCutTree(int n) : info
       (n + 1), tag(n + 1), ch(n + 1), p(n + 1), rev(n + 1) {}
  bool isrt(int x) {
    return ch[p[x]][0] != x && ch[p[x]][1] != x;
  }
  int pos(int x) { // x 是其 par 的左/右
    return ch[p[x]][1] == x;
  }
  void applyRev(int x) {
    if (!x) return;
    swap(ch[x][0], ch[x][1]);
    rev[x] ^= 1;
  }
  void apply(int x, const Tag &t) {
    if (!x) return;
    info[x].apply(t);
    tag[x].apply(t);
  }
  void push(int x) {
    if (rev[x]) {
      applyRev(ch[x][0]);
      applyRev(ch[x][1]);
      rev[x] = 0;
    }
    apply(ch[x][0], tag[x]);
    apply(ch[x][1], tag[x]);
    tag[x] = Tag();
  }
  void pull(int x) {
    if (!x) return;
    info[x].pull(info[ch[x][0]], info[ch[x][1]]);
  }
  void pushAll(int x) {
    if (!isrt(x)) pushAll(p[x]);
    push(x);
  }
  void rotate(int x) { // x 與其 par 交換位置
    int f = p[x], r = pos(x);
    ch[f][r] = ch[x][!r];
    if (ch[x][!r]) p[ch[x][!r]] = f;
    p[x] = p[f];
    if (!isrt(f)) ch[p[f]][pos(f)] = x;
    ch[x][!r] = f, p[f] = x;
    pull(f), pull(x);
  }
  void splay(int x) { // x 旋轉到當前的根
    pushAll(x);
    for (int f = p[x]; f = p[x], !isrt(x); rotate(x))
      if (!isrt(f)) rotate(pos(x) == pos(f) ? f : x);
  }
  // access(x), access(y) 可以回傳 LCA
  int access(int x) { // 根到 x 換成實鏈
    int c;
    for (c = 0; x; c = x, x = p[x]) {
      splay(x);
      ch[x][1] = c;
      pull(x);
    }
    return c;
  }
  void makeRoot(int x) { // x 變成所在樹的根
    access(x), splay(x), applyRev(x);
  }
  int findRoot(int x) {
    access(x), splay(x);
    while (ch[x][0]) x = ch[x][0];
    splay(x); return x;
  }
  void split(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(x), access(rt), splay(rt);
  }
  void link(int rt, int x) { // 以 rt 為根, x 為子樹根
    makeRoot(rt);
    access(x), splay(x);
    p[rt] = x;
    pull(x);
  }
  void cut(int rt, int x) { // 以 rt 為根, x 為子樹根
    split(rt, x);
    ch[rt][0] = p[x] = 0;
    pull(rt);
  }
  bool connected(int x, int y) {
    return findRoot(x) == findRoot(y);
  }
  bool neighbor(int x, int y) {
    if (!connected(x, y)) return false;
    split(x, y);
    return info[x].siz == 2; // psiz
  }
  void modify(int x, const Info &i) {
    access(x), splay(x);
    info[x] = i, pull(x);
  }
  void pathApply(int x, int y, const Tag &t) {
    assert(connected(x, y));
    split(x, y), apply(x, t);
  }
  Info pathQuery(int x, int y) {
    assert(connected(x, y));
    split(x, y); return info[x];
  }
};
```

3.8 LCT Extension [d41d8c]

```
struct Info {
  int siz = 0, vsiz
        = 0, psiz = 0; // 真實子樹大小, 虛子樹大小, 實鏈長度
  ll val = 0, sum = 0, vsum = 0, psum = 0;
  void apply(const Tag &t) {
    val = (val * t.mul % Mod + t.add) % Mod;
    sum = (sum - psum + Mod) % Mod;
    psum = (psum * t.mul % Mod + t.add * psiz % Mod) % Mod;
    sum = (sum + psum) % Mod;
  }
  void pull(const Info &l, const Info &r) {
    psiz = 1 + l.psiz + r.psiz;
    siz = 1 + l.siz + r.siz + vsiz;
    psum = (l.psum + r.psum + val) % Mod;
    sum = (l.sum + r.sum + val + vsum) % Mod;
  }
  void modify(const Info &i) { // 才不會蓋掉 vsum, vsiz
    val = i.val;
  } // 下面的 modify 也要改，不能直接 =
  void operator+=(const Info &i) { // 將實邊轉虛邊時加上資訊
    vsiz += i.siz;
    vsum = (vsum + i.sum) % Mod;
  }
  void operator-=(const Info &i) { // 將虛邊轉實邊時扣除資訊
    vsiz -= i.siz;
    vsum = (vsum - i.sum % Mod + Mod) % Mod;
  }
  Info asSubtree
      () const { // 當前節點作為 Virtual Subtree 根時的狀態
    Info res;
    res.siz = vsiz + 1;
    res.sum = (vsum + val) % Mod;
    return res;
  }
};
struct LinkCutTree { // 加在註解後面
  int access(int x) { // ...splay(x);
    if (ch[x][1]) info[x] += info[ch[x][1]]; // 原右子樹轉虛邊
    if (c) info[x] -= info[c];                // 虛邊轉新右子樹
  }
  void link(int rt, int x) { // ...p[rt] = x;
    info[x] += info[rt]; // rt 成為 x 的虛子樹
  }
  Info subtreeQuery
      (int rt, int x) { // 以 rt 當 root, x 的 subtree
    assert(connected(rt, x));
    makeRoot(rt);
    access(x), splay(x);
    return info[x].asSubtree();
  }
  Info componentQuery(int x) {
    makeRoot(x);
    return info[x];
  }
};
```

3.9 Treap [d41d8c]

```
template<class Info, class Tag = bool()>
struct Treap { // 0 -> initial root
  vector<Info> info;
  // vector<Tag> tag;
```

```cpp
  vector<int> siz, par, rev, pri;
  vector<array<int, 2>> ch;
  Treap(int n) : info(n + 1), siz
       (n + 1), par(n + 1), rev(n + 1), pri(n + 1), ch(n + 1) {
    // tag.resize(n + 1);
    for (int i = 1; i <= n; i++)
      siz[i] = 1, pri[i] = gen();
  }
  // void apply(int t, const Tag &v) {
  //     info[t].apply(siz[t], v);
  //     tag[t].apply(v);
  // }
  void push(int t) {
    if (rev[t]) {
      swap(ch[t][0], ch[t][1]);
      if (ch[t][0]) rev[ch[t][0]] ^= 1;
      if (ch[t][1]) rev[ch[t][1]] ^= 1;
      rev[t] = 0;
    }
    // apply(ch[t][0], tag[t]);
    // apply(ch[t][1], tag[t]);
    // tag[t] = Tag();
  }
  void pull(int t) {
    siz[t] = 1 + siz[ch[t][0]] + siz[ch[t][1]];
    info[t].pull(info[ch[t][0]], info[ch[t][1]]);
  }
  int merge(int a, int b) {
    if (!a || !b) return a ? a : b;
    push(a), push(b);
    if (pri[a] > pri[b]) {
      ch[a][1] = merge(ch[a][1], b);
      pull(a); return a;
    } else {
      ch[b][0] = merge(a, ch[b][0]);
      pull(b); return b;
    }
  }
  pair<int, int> split(int t, int k) {
    if (!t) return {0, 0};
    push(t);
    if (siz[ch[t][0]] >= k) {
      auto [a, b] = split(ch[t][0], k);
      ch[t][0] = b, pull(t);
      return {a, t};
    } else {
      auto [a, b] = split(ch[t][1], k - siz[ch[t][0]] - 1);
      ch[t][1] = a, pull(t);
      return {t, b};
    }
  }
  template<class F>   // 尋找區間內，第一個符合條件的
  int findFirst(int t, F &&pred) {
    if (!t) return 0;
    push(t);
    if (!pred(info[t])) return 0;
    int idx = findFirst(ch[t][0], pred);
    if (!idx
        ) idx = 1 + siz[ch[t][0]] + findFirst(ch[t][1], pred);
    return idx;
  }
  int getPos(int rt, int t) { // get t's index in array
    int res = siz[t] + 1;
    while (t != rt) {
      int p = par[t];
      if (ch[p][1] == t) res += siz[ch[p][0]] + 1;
      t = p;
    }
    return res;
  }
  void getArray(int t, vector<Info> &a) {
    if (!t) return;
    push(t);
    getArray(ch[t][0], a);
    a.push_back(info[t]);
    getArray(ch[t][1], a);
  }
};
struct Tag {
  int setVal; ll add;
  void apply(const Tag &t) {
    if (t.setVal) {
      setVal = t.setVal;
      add = t.add;
    } else {
      add += t.add;
    }
  }
};
struct Info {
  ll val, sum;
  void apply(int siz, const Tag &t) {
    if (t.setVal) {
      val = t.setVal;
      sum = 1LL * siz * t.setVal;
    }
    val += t.add;
    sum += 1LL * siz * t.add;
  }
  void pull(const Info &l, const Info &r) {
    sum = val + l.sum + r.sum;
  }
};
```

3.10 KDTree [d41d8c]

```cpp
struct Info {
  static constexpr int DIM = 2;
  array<int, DIM> x, L, R;
  int v = 0, sum = 0;
  void pull(const Info &l, const Info &r) {
    sum = v + l.sum + r.sum;
  }
};
struct KDTree {
  static constexpr int DIM = Info::DIM;
  vector<Info> info;
  vector<int> rt, l, r, p;
  int n, lg;
  KDTree(int n) : n(n),
        lg(__lg(n)), info(1), rt(lg + 1), l(n + 1), r(n + 1) {}
  void pull(int p) {
    info[p].L = info[p].R = info[p].x;
    info[p].pull(info[l[p]], info[r[p]]);
    for (int ch : {l[p], r[p]}) {
      if (!ch) continue;
      for (int k = 0; k < DIM; k++) {
        info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
        info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
      }
    }
  }
  int rebuild(int l, int r, int dep = 0) {
    if (r == l) return 0;
    int m = (l + r) / 2;
    nth_element(p.begin() + l, p.begin() + m, p.begin() + r,
      [&](int x,
          int y) { return info[x].x[dep] < info[y].x[dep]; });
    int x = p[m];
    this->l[x] = rebuild(l, m, (dep + 1) % DIM);
    this->r[x] = rebuild(m + 1, r, (dep + 1) % DIM);
    pull(x);
    return x;
  }
  void append(int &x) {
    if (!x) return;
    p.push_back(x);
    append(l[x]);
    append(r[x]);
    x = 0;
  }
  void addNode(const Info &i) {
    p.assign(1, info.size());
    info.push_back(i);
    for (int j = 0;; j++) {
      if (!rt[j]) {
        rt[j] = rebuild(0, p.size());
        break;
      } else {
        append(rt[j]);
      }
    }
  }
  Info query(int
        p, const array<int, DIM> &l, const array<int, DIM> &r) {
    if (!p) return Info();
    bool inside = true;
    for (int k = 0; k < DIM; k++) {
      inside &= (l[k] <= info[p].L[k] && info[p].R[k] <= r[k]);
    }
    if (inside) return info[p];
    for (int k = 0; k < DIM; k++) {
      if (info[p].R[k] < l[k] || r[k] < info[p].L[k]) {
        return Info();
      }
    }
    Info ans;
    inside = true;
    for (int k = 0; k < DIM; k++) {
      inside &= l[k] <= info[p].x[k] && info[p].x[k] <= r[k];
    }
    if (inside) ans = info[p];
    ans.pull(query(this->l[p], l, r), query(this->r[p], l, r));
    return ans;
  }
  Info query
        (const array<int, DIM> &l, const array<int, DIM> &r) {
    Info res;
    for (int i = 0; i <= lg; i++) {
      res.pull(res, query(rt[i], l, r));
    }
    return res;
  }
};
```

3.11 RMQ [d41d8c]

```cpp
template<class T, class F = less<T>> struct RMQ { // [l, r)
  int n;
  F cmp = F();
  vector<vector<T>> g;
  RMQ() {}
  RMQ(const vector<T> &a, F cmp = F()) : cmp(cmp) {
    init(a);
```

```cpp
  }
  void init(const vector<T> &a) {
    n = a.size();
    if (!n) return;
    int lg = __lg(n);
    g.resize(lg + 1);
    g[0] = a;
    for (int j = 1; j <= lg; j++) {
      g[j].resize(n - (1 << j) + 1);
      for (int i = 0; i <= n - (1 << j); i++)
        g[j][i] = min
             (g[j - 1][i], g[j - 1][i + (1 << (j - 1))], cmp);
    }
  }
  T operator()(int l, int r) {
    assert(0 <= l && l < r && r <= n);
    int lg = __lg(r - l);
    return min(g[lg][l], g[lg][r - (1 << lg)], cmp);
  }
};
```

3.12 Mo [d41d8c]

```cpp
struct Query { int id, l, r; };
void mo(vector<Query> &q) {
  int blk = sqrt(q.size());
  sort(q
       .begin(), q.end(), [&](const Query &a, const Query &b) {
    int x = a.l / blk, y = b.l / blk;
    return x == y ? a.r < b.r : x < y;
  });
}
int nl = 0, nr = -1;
for (auto [id, l, r] : qry) {
  while (nr < r) nr++, addR();
  while (l < nl) nl--, addL();
  while (r < nr) delR(), nr--;
  while (nl < l) delL(), nl++;
}
```

4 Flow Matching

4.1 Dinic [d41d8c]

```cpp
template<class T> struct Dinic {
  // argument time: O(VE), O(E) for unit capacity,
  // argument number: O(V), min(O(E^0.5), O(V^2/3)) for
       unit capacity, O(V^0.5) for deg_in(u) or deg_out(u) <= 1
  // so bipartite matching: O(EV^0.5)
  struct Edge {
    int to;
    T f, cap; // 流量跟容量
  };
  int n, m, s, t;
  const T INF_FlOW = numeric_limits<T>::max() / 2;
  vector<vector<int>> g;
  vector<Edge> e;
  vector<int> h, cur;
  Dinic(int n) : n(n), m(0), g(n), h(n), cur(n) {}
  void addEdge(int u, int v, T cap) {
    e.push_back({v, 0, cap});
    e.push_back({u, 0, 0});
    g[u].push_back(m++);
    g[v].push_back(m++);
  }
  bool bfs() {
    fill(h.begin(), h.end(), -1);
    h[s] = 0; queue<int> q;
    q.push(s);
    while (!q.empty()) {
      int u = q.front(); q.pop();
      for (int id : g[u]) {
        auto [v, f, cap] = e[id];
        if (f != cap && h[v] == -1) {
          h[v] = h[u] + 1;
          if (v == t) return true;
          q.push(v);
        }
      }
    }
    return false;
  }
  T dfs(int u, T flow) {
    if (flow == 0) return 0;
    if (u == t) return flow;
    for (int &i = cur[u]; i < g[u].size(); i++) {
      int j = g[u][i];
      auto [v, f, cap] = e[j];
      if (f == cap || h[u] + 1 != h[v]) continue;
      T mn = dfs(v, min(flow, cap - f));
      if (mn > 0) {
        e[j].f += mn;
        e[j ^ 1].f -= mn;
        return mn;
      }
    }
    return T{};
  }
  T work(int s_, int t_) {
    s = s_; t = t_; T f = 0;
    while (bfs()) {
      fill(cur.begin(), cur.end(), 0);
      while (true) {
        T res = dfs(s, INF_FlOW);
        if (res == 0) break;
        f += res;
      }
    }
    return f;
  }
  void reuse(int n_) { // 走殘留網路，res += f
    while (n < n_) {
      g.emplace_back();
      h.emplace_back();
      cur.emplace_back();
      n += 1;
    }
  }
};
```

4.2 Min Cut [d41d8c]

```cpp
void policeChase(int n, Dinic<int> &d) {
  int ans = d.work(0, n - 1);
  vector<int> vis(n);
  [&](this auto &&self, int u) -> void {
    if (vis[u]) return;
    vis[u] = 1;
    for (int id : d.g[u]) {
      auto [to, f, cap] = d.e[id];
      if (cap - f > 0) self(to);
    }
  } (0);
  for (int i = 0; i < n; i++) {
    if (!vis[i]) continue;
    for (int id : d.g[i]) {
      if (id & 1) continue;
      auto e = d.e[id];
      if (!vis[e.to]) cout << i + 1 << " " << e.to + 1 << "\n";
    }
  }
}
```

4.3 MCMF [d41d8c]

```cpp
template<class Tf, class Tc> struct MCMF {
  struct Edge {
    int to;
    Tf f, cap; // 流量跟容量
    Tc cost;
  };
  int n, m, s, t;
  const Tf INF_FLOW = numeric_limits<Tf>::max() / 2;
  const Tc INF_COST = numeric_limits<Tc>::max() / 2;
  vector<Edge> e;
  vector<vector<int>> g;
  vector<Tc> dis, pot;
  vector<int> rt, inq;
  MCMF(int n) : n(n), m(0), g(n) {}
  void addEdge(int u, int v, Tf cap, Tc cost) {
    e.push_back({v, 0, cap, cost});
    e.push_back({u, 0, 0, -cost});
    g[u].push_back(m++);
    g[v].push_back(m++);
  }
  bool spfa() {   // O(FVE)
    dis.assign(n, INF_COST);
    rt.assign(n, -1), inq.assign(n, 0);
    queue<int> q; q.push(s);
    dis[s] = 0, inq[s] = 1;
    while (!q.empty()) {
      int u = q.front(); q.pop();
      inq[u] = 0;
      for (int id : g[u]) {
        auto [v, f, cap, cost] = e[id];
        Tc ndis = dis[u] + cost + pot[u] - pot[v];
        if (f < cap && dis[v] > ndis) {
          dis[v] = ndis, rt[v] = id;
          if (!inq[v])
            q.push(v), inq[v] = 1;
        }
      }
    }
    return dis[t] != INF_COST;
  }
  bool dijkstra() {   // O(FElogV)
    dis.assign(n, INF_COST), rt.assign(n, -1);
    priority_queue<pair<Tc, int
        >, vector<pair<Tc, int>>, greater<pair<Tc, int>>> pq;
    dis[s] = 0; pq.emplace(dis[s], s);
    while (!pq.empty()) {
      auto [d, u] = pq.top(); pq.pop();
      if (dis[u] < d) continue;
      for (int id : g[u]) {
        auto [v, f, cap, cost] = e[id];
        Tc ndis = dis[u] + cost + pot[u] - pot[v];
        if (f < cap && dis[v] > ndis) {
          dis[v] = ndis, rt[v] = id;
          pq.emplace(ndis, v);
        }
      }
    }
    return dis[t] != INF_COST;
```

```
  }
  pair<Tf, Tc> work(int s_, int t_, Tf need) {
    s = s_, t = t_; pot.assign(n, 0);
    Tf flow{}; Tc cost{}; int fr = 0;
    while (fr++ ? dijkstra() : spfa()) {
      for (int i = 0; i < n; i++)
        dis[i] += pot[i] - pot[s];
      Tf f = need;
      for (int i = t; i != s; i = e[rt[i] ^ 1].to)
        f = min(f, e[rt[i]].cap - e[rt[i]].f);
      for (int i = t; i != s; i = e[rt[i] ^ 1].to)
        e[rt[i]].f += f, e[rt[i] ^ 1].f -= f;
      flow += f, need -= f;
      cost += f * dis[t];
      swap(dis, pot);
      if (need == 0) break;
    }
    return {flow, cost};
  }
  void reset() {
    for (int i = 0; i < m; i++) e[i].f = 0;
  }
};
```

4.4 Hungarian [d41d8c]

```
struct Hungarian { // 0-based, O(VE)
  int n, m;
  vector<vector<int>> adj;
  vector<int> used, vis;
  vector<pair<int, int>> match;
  Hungarian(int n, int m) : n(n), m(m) {
    adj.assign(n + m, {});
    used.assign(n + m, -1);
    vis.assign(n + m, 0);
  }
  void addEdge(int u, int v) {
    adj[u].push_back(n + v);
    adj[n + v].push_back(u);
  }
  bool dfs(int u) {
    int sz = adj[u].size();
    for (int i = 0; i < sz; i++) {
      int v = adj[u][i];
      if (vis[v] == 0) {
        vis[v] = 1;
        if (used[v] == -1 || dfs(used[v])) {
          used[v] = u;
          return true;
        }
      }
    }
    return false;
  }
  vector<pair<int, int>> work() {
    match.clear();
    used.assign(n + m, -1);
    vis.assign(n + m, 0);
    for (int i = 0; i < n; i++) {
      fill(vis.begin(), vis.end(), 0);
      dfs(i);
    }
    for (int i = n; i < n + m; i++)
      if (used[i] != -1)
        match.emplace_back(used[i], i - n);
    return match;
  }
};
```

- 有向無環圖:
  - 最小不相交路徑覆蓋: 最小路徑數 = 頂點數–最大匹配數
  - 最小相交路徑覆蓋: 先用 Floyd 求傳遞封包，有連邊就建邊，然後再套最小不相交路徑覆蓋
- 二分圖:
  - 最小點覆蓋: 選出一些點，使每條邊至少有一個端點被選到的最少數量
  - 最小點覆蓋 = 最大匹配數
  - 還原解: flow 的作法是從源點開始 dfs，只走 cap - flow > 0 的邊，最後挑選左邊還沒被跑過的點和右邊被跑過的點當作覆蓋的點
  - 最大獨立集: 選出一些點，使這些點兩兩沒有邊連接的最大數量
  - 最大獨立集 = 點數–最大匹配數
  - 還原解: 最小點覆蓋的補集
  - 最少邊覆蓋: 選出一些邊，讓所有點都覆蓋到的最少數量
  - 最少邊覆蓋 = 點數–孤立點數 - 最大匹配數
  - 還原解: 最大匹配 + 未被匹配的非孤立點隨便配

5 String

5.1 Hash [234076]

```
const int D = 59;
vector<int> rollingHash(string &s) {
  vector<int> a {0};
  for (auto c : s)
    a.push_back(mul(a.back(), D) + (c - 'A' + 1));
  return a;
}
int qryHash(vector<int> &h, int l, int r) { // [l, r)
  return sub(h[r], mul(h[l], power(D, r - l)));
}
```

5.2 KMP [cdf28b]

```
template<class T> struct KMP {
  int n;
  T pat;
  vector<int> f;
  // f 存匹配失敗時，移去哪
  // 也就是 pat[0, i] 的最長共同前後綴長度
  // ex : a   b   c a b c
  //      -1 -1 -1 0 1 2
  KMP(const T &pat) : n(pat.size()), pat(pat), f(n, -1) {
    for (int i = 1; i < n; i++) {
      int j = f[i - 1];
      while (j != -1 && pat[j + 1] != pat[i]) j = f[j];
      if (pat[j + 1] == pat[i]) f[i] = j + 1;
    }
  }
  vector<int> work(const T &s) {
    vector<int> pos;
    for (int i = 0, j = -1; i < s.size(); i++) {
      while (j != -1 && pat[j + 1] != s[i]) j = f[j];
      if (pat[j + 1] == s[i]) j++;
      if (j + 1 == n) {
        pos.push_back(i - j);
        j = f[j];
      }
    }
    return pos;
  }
};
```

5.3 Z Function [a1f65d]

```
// z[i] 表示 s 和 s[i, n - 1] (以 s[i] 開頭的後綴)
// 的最長公共前綴 (LCP) 的長度
template<class T> vector<int> Z(const T &s) {
  int n = s.size();
  vector<int> z(n);
  z[0] = n; // lcp(s, s), -1 or n
  for (int i = 1, j = 1; i < n; i++) {
    z[i] = max(0, min(j + z[j] - i, z[i - j]));
    while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
    if (i + z[i] > j + z[j]) j = i;
  }
  return z;
}
```

5.4 Manacher [9f76ee]

```
// 找到對於每個位置的迴文半徑
template<class T> vector<int> manacher(const T &s) {
  vector<int> t {-1};
  for (auto c : s) t.push_back(c), t.push_back(-1);
  int n = t.size();
  vector<int> r(n);
  for (int i = 0,
       j = 0; i < n; i++) { // i 是中心, j 是最長回文字串中心
    if (2 * j - i >= 0 && j + r[j] > i)
      r[i] = min(r[2 * j - i], j + r[j] - i);
    while (i - r[
        i] >= 0 && i + r[i] < n && t[i - r[i]] == t[i + r[i]])
      r[i]++;
    if (i + r[i] > j + r[j]) j = i;
  }
  return r;
}
// # a # b # a #
// 1 2 1 4 1 2 1
// # a # b # b # a #
// 1 2 1 2 5 2 1 2 1
// 值 -1 代表原回文字串長度
// (id - val + 1) / 2 可得原字串回文開頭
```

5.5 Trie [2f0f82]

```
const int N = 1E7; // 0 -> initial state
const int ALPHABET_SIZE = 26;
int tot = 0;
int trie[N][ALPHABET_SIZE], cnt[N];
void reset() {
  tot = 0, fill_n(trie[0], ALPHABET_SIZE, 0);
}
int newNode() {
  int x = ++tot;
  cnt[x] = 0, fill_n(trie[x], ALPHABET_SIZE, 0);
  return x;
}
void add(const string &s, int i = 0, int p = 0) {
  if (i == s.size()) {
    cnt[p]++;
    return;
  }
  int &q = trie[p][s[i] - 'a'];
  if (!q) q = newNode();
  add(s, i + 1, q);
}
int find(const string &s, int i = 0, int p = 0) {
  if (i == s.size()) return cnt[p];
  int q = trie[p][s[i] - 'a'];
  if (!q) return 0;
  return find(s, i + 1, q);
}
```

5.6 SA [9262af]

```cpp
template<class T> struct SuffixArray {
  int n;
  vector<int> sa, rk, lc;
  // n: 字串長度
  // sa: 後綴數組，sa[i] 表示第 i 小的後綴的起始位置
  // rk: 排名數組，rk[i] 表示從位置 i 開始的後綴的排名
  // lc: LCP
       數組，lc[i] 表示 sa[i] 和 sa[i + 1] 的最長公共前綴長度
  SuffixArray
      (const T &s) : n(s.size()), sa(n), lc(n - 1), rk(n) {
    iota(sa.begin(), sa.end(), 0);
    sort(sa.begin(),
        sa.end(), [&](int a, int b) { return s[a] < s[b]; });
    rk[sa[0]] = 0;
    for (int i = 1; i < n; i++)
      rk[sa[i]] = rk[sa[i - 1]] + (s[sa[i]] != s[sa[i - 1]]);
    int k = 1;
    vector<int> tmp, cnt(n);
    tmp.reserve(n);
    while (rk[sa[n - 1]] < n - 1) {
      tmp.clear();
      for (int i = 0; i < k; i++) tmp.push_back(n - k + i);
      for (auto i : sa) if (i >= k) tmp.push_back(i - k);
      fill(cnt.begin(), cnt.end(), 0);
      for (int i = 0; i < n; i++) cnt[rk[i]]++;
      for (int i = 1; i < n; i++) cnt[i] += cnt[i - 1];
      for (int i =
          n - 1; i >= 0; i--) sa[--cnt[rk[tmp[i]]]] = tmp[i];
      swap(rk, tmp); rk[sa[0]] = 0;
      for (int i = 1; i < n; i++) rk[sa[i]] = rk[
          sa[i - 1]] + (tmp[sa[i - 1]] < tmp[sa[i]] || sa[i -
          1] + k == n || tmp[sa[i - 1] + k] < tmp[sa[i] + k]);
      k *= 2;
    }
    for (int i = 0, j = 0; i < n; i++) {
      if (rk[i] == 0) {
        j = 0;
      } else {
        for (j -= j > 0; i + j < n && sa[rk[i] - 1]
            + j < n && s[i + j] == s[sa[rk[i] - 1] + j]; j++);
        lc[rk[i] - 1] = j;
      }
    }
  }
};
RMQ<int> rmq(sa.lc);
auto lcp = [&](int i, int j) { // [i, j]
  i = sa.rk[i], j = sa.rk[j];
  if (i > j) swap(i, j);
  assert(i != j);
  return rmq(i, j);
};
```

5.7 AC [d34486]

```cpp
struct AC {
  static constexpr int ALPHABET_SIZE = 26;
  struct Node {
    int fail; // 指向最長後綴
    int cnt; // 有多少模式字串是自己的後綴
    array<int, ALPHABET_SIZE> ch, next;
    // next 是補全後的轉移
  };
  vector<Node> t;
  AC() : t(1) {}
  int newNode() {
    t.emplace_back();
    return t.size() - 1;
  }
  int insert(const string &s) {
    int u = 0;
    for (char c : s) {
      if (!t[u].ch[c - 'a']) {
        int v = newNode();
        t[u].ch[c - 'a'] = v;
      }
      u = t[u].ch[c - 'a'];
    }
    t[u].cnt++;
    return u;
  }
  void build() {
    queue<int> q;
    for (int c = 0; c < ALPHABET_SIZE; c++) {
      if (t[0].ch[c]) {
        q.push(t[0].ch[c]);
        t[0].next[c] = t[0].ch[c];
      }
    }
    while (!q.empty()) {
      int u = q.front(); q.pop();
      for (int c = 0; c < ALPHABET_SIZE; c++) {
        if (t[u].ch[c]) {
          int v = t[u].ch[c];
          t[v].fail = t[t[u].fail].next[c];
          t[v].cnt += t[t[v].fail].cnt;
          t[u].next[c] = v;
          q.push(v);
        } else {
          t[u].next[c] = t[t[u].fail].next[c];
        }
      }
    }
  }
};
```

5.8 SAM [83a519]

```cpp
struct SAM {
  // 0 -> initial state
  static constexpr int ALPHABET_SIZE = 26;
  // node -> strings with the same endpos set
  // link -> longest suffix with different endpos set
  // len -> state's longest suffix
  // fpos -> first endpos
  // strlen range -> [len(link) + 1, len]
  struct Node {
    int len, link = -1, fpos;
    array<int, ALPHABET_SIZE> next;
  };
  vector<Node> t;
  SAM() : t(1) {}
  int newNode() {
    t.emplace_back();
    return t.size() - 1;
  }
  int extend(int p, int c) {
    int cur = newNode();
    t[cur].len = t[p].len + 1;
    t[cur].fpos = t[cur].len - 1;
    while (p != -1 && !t[p].next[c]) {
      t[p].next[c] = cur;
      p = t[p].link;
    }
    if (p == -1) {
      t[cur].link = 0;
    } else {
      int q = t[p].next[c];
      if (t[p].len + 1 == t[q].len) {
        t[cur].link = q;
      } else {
        int r = newNode();
        t[r] = t[q];
        t[r].len = t[p].len + 1;
        while (p != -1 && t[p].next[c] == q) {
          t[p].next[c] = r;
          p = t[p].link;
        }
        t[q].link = t[cur].link = r;
      }
    }
    return cur;
  }
};
void solve(int n, string s, ll k) { // Substring Order II
  vector<int> last(n + 1);
  SAM sam;
  for (int i = 0; i < n; i++)
    last[i + 1] = sam.extend(last[i], s[i] - 'a');
  int sz = sam.t.size();

  vector<int> cnt(sz); // endpos size
  for (int i = 1; i <= n; i++) cnt[last[i]]++;
  vector<vector<int>> g(sz);
  for (int i = 1; i < sz; i++)
    g[sam.t[i].link].push_back(i);
  [&](this auto &&self, int u) -> void {
    for (auto v : g[u]) self(v), cnt[u] += cnt[v];
  }; (0);

  vector<ll> dp(sz, -1);
  // for any path from
       root, how many substring's prefix is the the path string
  [&](this auto &&self, int u) -> ll {
    if (dp[u] != -1) return dp[u];
    dp[u] = cnt[u]; // distinct: = 1
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
      int v = sam.t[u].next[c];
      if (v) dp[u] += self(v);
    }
    return dp[u];
  } (0);

  int p = 0; string ans;
  while (k > 0) { // 1-based
    for (int c = 0; c < SAM::ALPHABET_SIZE; c++) {
      int v = sam.t[p].next[c];
      if (v) {
        if (k > dp[v]) {
          k -= dp[v];
        } else {
          ans.push_back('a' + c);
          k -= cnt[v]; // distinct: --
          p = v; break;
        }
      }
    }
  } cout << ans << "\n";
}
```

5.9 Palindrome Tree [e5a1ed]

```
struct PAM {
  // 0 -> even root, 1 -> odd root
  static constexpr int ALPHABET_SIZE = 26;
  // fail -> longest prefix(suffix) palindrome
  // number end at i = end at link[last[i]] + 1
  struct Node {
    int len, fail, cnt;
    array<int, ALPHABET_SIZE> next;
    Node() : len{}, fail{}, next{} {}
  };
  vector<int> s;
  vector<Node> t;
  PAM() {
    t.assign(2, Node());
    t[0].len = 0, t[0].fail = 1;
    t[1].len = -1;
  }
  int newNode() {
    t.emplace_back();
    return t.size() - 1;
  }
  int getFail(int p, int i) {
    while (i - t[p].len < 1 || s[i - t[p].len - 1] != s[i])
      p = t[p].fail;
    return p;
  }
  int extend(int p, int c) {
    int i = s.size();
    s.push_back(c);
    p = getFail(p, i);
    if (!t[p].next[c]) {
      int r = newNode();
      int v = getFail(t[p].fail, i);
      t[r].len = t[p].len + 2;
      t[r].fail = t[v].next[c];
      t[p].next[c] = r;
    }
    return p = t[p].next[c];
  }
};
void solve() {
  string s; cin >> s;
  int n = s.length();
  vector<int> last(n + 1);
  last[0] = 1;
  PAM pam;
  for (int i = 0; i < n; i++)
    last[i + 1] = pam.extend(last[i], s[i] - 'a');
  int sz = pam.t.size();
  vector<int> cnt(sz);
  for (int i = 1; i <= n; i++)
    cnt[last[i]]++; // 去重 = 1
  for (int i = sz - 1; i > 1; i--)
    cnt[pam.t[i].fail] += cnt[i];
}
```

5.10 Duval [f0a5c2]

```
// duval_algorithm
// 將字串分解成若干個非嚴格遞減的非嚴格遞增字串
template<class T> vector<T> duval(const T &s) {
  int i = 0, n = s.size();
  vector<T> res;
  while (i < n) {
    int k = i, j = i + 1;
    for (; j < n && s[k] <= s[j]; j++) {
      if (s[k] < s[j]) k = i;
      else k++;
    }
    while (i <= k) {
      res.emplace_back(s.begin() + i, s.begin() + i + j - k);
      i += j - k;
    }
  }
  return res;
}
// 最小旋轉字串
template<class T> T minRound(T s) {
  s.insert(s.end(), s.begin(), s.end());
  int i = 0, n = s.size(), start = i;
  while (i < n / 2) {
    start = i;
    int k = i, j = i + 1;
    for (; j < n && s[k] <= s[j]; j++) {
      if (s[k] < s[j]) k = i;
      else k++;
    }
    while (i <= k) i += j - k;
  }
  return {s.begin() + start, s.begin() + start + n / 2};
}
```

6 Math

6.1 Mint [3c599b]

```
const int P = 1E9 + 7;
ll mul(ll a, ll b, ll p) { // P 超過 int 再用，慢
  ll res = a * b - ll(1.L * a * b / p) * p;
  res %= p;
  if (res < 0) res += p;
  return res;
}
template<class T> constexpr T power(T a, ll b) {
  T res{1};
  for (; b > 0; b >>= 1, a = a * a)
  if (b & 1) res = res * a;
  return res;
}
template<int P> struct Mint {
  static int Mod;
  static int getMod() { return P > 0 ? P : Mod; }
  static void setMod(int Mod_) { Mod = Mod_; }
  ll x;
  Mint(ll v = 0) {
    x = v % getMod();
    if (x < 0) x += getMod();
  }
  explicit operator ll() const { return x; }
  Mint operator-() const { return getMod() - x; }
  Mint inv() const { return power(*this, getMod() - 2); }
  Mint operator+(Mint a) const { return x + a.x; }
  Mint operator-(Mint a) const { return x - a.x; }
  Mint operator*(Mint a) const { return x * a.x; }
  Mint operator/(Mint a) const { return *this * a.inv(); }

  Mint &operator+=(Mint a) { return *this = *this + a; }
  Mint &operator-=(Mint a) { return *this = *this - a; }
  Mint &operator*=(Mint a) { return *this = *this * a; }
  Mint &operator/=(Mint a) { return *this = *this / a; }

  bool operator==(Mint y) const { return x == y.x; }
  bool operator!=(Mint y) const { return x != y.x; }

  friend istream &operator>>(istream &is, Mint &a)
  { ll v; is >> v; a = Mint(v); return is; }
  friend ostream &operator<<(ostream &os, Mint a)
  { return os << a.x; }
};
template<> int Mint<0>::Mod = 998244353;
using Z = Mint<P>;

vector<Z> fac, invfac, inv;
void init(int n) {
  fac.resize(n + 1);
  invfac.resize(n + 1);
  inv.resize(n + 1);
  fac[0] = invfac[0] = 1;
  for (int i = 1; i <= n; i++) {
    fac[i] = fac[i - 1] * i;
  }
  invfac[n] = fac[n].inv();
  for (int i = n; i > 0; i--) {
    invfac[i - 1] = invfac[i] * i;
    inv[i] = invfac[i] * fac[i - 1];
  }
}
Z binom(int n, int m) {
  if (n < m || m < 0) return 0;
  return fac[n] * invfac[m] * invfac[n - m];
}
Z lucas(ll n, ll m) { // O(p + T log(n)), p is prime
  return m ? binom(n % P, m % P) * lucas(n / P, m / P) : 1;
}
```

6.2 Sieve [83c88a]

```
vector<int> minp, primes;
vector<int> phi, mu, pnum; // 質因數種類數
vector<int> mpnum, dnum; // 最小質因數的冪次數, 約數數量
vector<int> powpref, dsum; // 約數和
// dmul[i] = i ^ (dnum[i] / 2) for dnum[i] even
// dmul[i] = k ^ dnum[i], k * k = i else
void sieve(int n) {
  minp.resize(n + 1);
  phi.resize(n + 1);
  mu.resize(n + 1);
  pnum.resize(n + 1);

  mpnum.resize(n + 1);
  dnum.resize(n + 1);

  powpref.resize(n + 1);
  dsum.resize(n + 1);

  phi[1] = mu[1] = 1;
  dnum[1] = 1;
  powpref[1] = dsum[1] = 1;
  for (int i = 2; i <= n; i++) {
    if (!minp[i]) {
      minp[i] = i;
      primes.push_back(i);

      phi[i] = i - 1;
      mu[i] = -1;
      pnum[i] = 1;

      mpnum[i] = 1;
      dnum[i] = 2;

      powpref[i] = i + 1;
```

```
      dsum[i] = i + 1;
    }
    for (int p : primes) {
      if (i * p > n) break;
      minp[i * p] = p;
      if (p == minp[i]) {
        phi[i * p] = phi[i] * p;
        mu[i * p] = 0;
        pnum[i * p] = pnum[i];

        mpnum[i * p] = mpnum[i] + 1;
        dnum[i *
             p] = dnum[i] / mpnum[i * p] * (mpnum[i * p] + 1);

        powpref[i * p] = powpref[i] * p + 1;
        dsum[i * p] = dsum[i] / powpref[i] * powpref[i * p];
        break;
        // i * p = (p * x) * p，就不篩 p < q 的 i * q 了
        // i * q = (p * x) * q = p * (x * q)
        // 到達 x * q 再用 p 篩掉就好
      } else {
        phi[i * p] = phi[i] * (p - 1);
        mu[i * p] = -mu[i];
        pnum[i * p] = pnum[i] + 1;

        mpnum[i * p] = 1;
        dnum[i * p] = dnum[i] * 2;

        powpref[i * p] = p + 1;
        dsum[i * p] = dsum[i] * (p + 1);
      }
    }
  }
}
```

6.3 Matrix [6b2cbc]

```
using Matrix = vector<vector<Z>>;
Matrix operator*(const Matrix &a, const Matrix &b) {
  int n = a.size(), k = a[0].size(), m = b[0].size();
  assert(k == b.size());
  Matrix res(n, vector<Z>(m));
  for (int i = 0; i < n; i++)
    for (int j = 0; j < m; j++)
      for (int l = 0; l < k; l++)
        res[i][j] += a[i][l] * b[l][j];
  return res;
}
Matrix power(Matrix a, ll b) {
  int n = a.size();
  Matrix res(n, vector<Z>(n));
  for (int i = 0; i < n; i++) res[i][i] = 1;
  for (; b > 0; b >>= 1, a = a * a)
    if (b & 1) res = res * a;
  return res;
}
```

6.4 Miller Rabin Pollard Rho [394cfb]

```
ll mul(ll a, ll b, ll p) {
  ll res = a * b - ll(1.L * a * b / p) * p;
  res %= p;
  if (res < 0) res += p;
  return res;
}
ll power(ll a, ll b, ll p) {
  ll res {1};
  for (; b; b /= 2, a = mul(a, a, p))
    if (b & 1) res = mul(res, a, p);
  return res;
}
vector<ll
    > chk {2, 325, 9375, 28178, 450775, 9780504, 1795265022};
bool check(ll a, ll d, int s, ll n) {
  a = power(a, d, n);
  if (a <= 1) return 1;
  for (int i = 0; i < s; i++, a = mul(a, a, n)) {
    if (a == 1) return 0;
    if (a == n - 1) return 1;
  }
  return 0;
}
bool isPrime(ll n) {
  if (n < 2) return 0;
  if (n % 2 == 0) return n == 2;
  ll d = n - 1, s = 0;
  while (d % 2 == 0) d /= 2, s++;
  for (ll i : chk)
    if (!check(i, d, s, n)) return 0;
  return 1;
}
const vector<ll> small = {2, 3, 5, 7, 11, 13, 17, 19};
ll findFactor(ll n) {
  if (isPrime(n)) return 1;
  for (ll p : small)
    if (n % p == 0) return p;
  ll x, y = 2, d, t = 1;
  auto f = [&](ll a) {
    return (mul(a, a, n) + t) % n;
  };
  for (int l = 2; ; l *= 2) {
    x = y;
    int m = min(l, 32);
    for (int i = 0; i < l; i += m) {
      d = 1;
      for (int j = 0; j < m; j++)
        y = f(y), d = mul(d, abs(x - y), n);
      ll g = __gcd(d, n);
      if (g == n) {
        l = 1, y = 2, ++t;
        break;
      }
      if (g != 1) return g;
    }
  }
}
map<ll, int> res;
void pollardRho(ll n) {
  if (n == 1) return;
  if (isPrime(n)) {
    res[n]++;
    return;
  }
  ll d = findFactor(n);
  pollardRho(n / d), pollardRho(d);
}
```

6.5 Primitive Root [b0ba96]

```
int findPrimitiveRoot(ll m) {
  Mlong<0>::setMod(m); // Mlong if needed
  ll phi = m; // m - 1 if m prime
  res.clear();
  pollardRho(m);
  for (auto [p, _] : res) phi = phi / p * (p - 1);
  vector<ll> pr; // prime factors of phi
  res.clear();
  pollardRho(phi);
  for (auto &[p, _] : res) pr.push_back(p);
  for (int i = 1; i <= m; i++) {
    bool ok = true;
    for (int j = 0; j < int(pr.size()) && ok; j++)
      ok &= power(Mlong<0>(i), phi / pr[j]).x != 1;
    if (ok) return i;
  }
  return -1; // m != 1, 2, 4, p^k, 2(p^k)
}
```

6.6 CRT [35109b]

```
// ax = b (mod m) 的前提是 gcd(a, m) | b
// Ax + By = C     (g = gcd(A, B))
// (A/g) * x + (B/g) * y = C/g
// solve (A/g) * x' + (B/g) * y' = 1
// x = (x' * (C/g)) + i(B/g)
pair<ll, ll> exgcd(ll a, ll b) {
  if (b == 0) return {1, 0};
  auto [y, x] = exgcd(b, a % b);
  return {x, y - (a / b) * x};
} // 結果可能為負，注意正規化
// Smallest non-negative solution
using i128 = __int128_t;
pair<ll, ll> CRT(ll r1, ll m1, ll r2, ll m2) {
  ll g = __gcd(m1, m2);
  if ((r2 - r1) % g) return {-1, g};
  m1 /= g, m2 /= g;
  auto [p1, p2] = exgcd(m1, m2);
  i128 lcm = i128(m1) * m2 * g;
  i128 res = i128(p1) * (r2 - r1) * m1 + r1;
  return {(res % lcm + lcm) % lcm, lcm};
}
pair<ll, ll> EXCRT(vector<pair<int, int>> a) {
  ll R = 0, M = 1;
  for (auto [r, m] : a) {
    auto [res, lcm] = CRT(R, M, r, m);
    if (res == -1) return {-1, -1};
    R = res, M = lcm;
  }
  return {R, M};
}
// gcd(mod) = 1, support 3 1E9 Mod
pair<i128, i128> CRT(vector<pair<int, int>> a) {
  i128 s = 1, res = 0;
  for (auto [r, m] : a) s *= m;
  for (auto [r, m] : a) {
    i128 t = s / m;
    res = (res + r * t % s * exgcd(t, m).first % s) % s;
  }
  return {(res + s) % s, s};
}
```

6.7 XOR Basis [02f0c0]

```
auto add = [&](vector<int> &bas, int x) {
  for (auto i : bas) x = min(x, x ^ i);
  if (x) bas.push_back(x);
};
sort(bas.begin(), bas.end()); // 最簡化列梯
for (auto i = bas.begin(); i != bas.end(); i++) {
  for (auto j = next(i); j != bas.end(); j++) {
    *j = min(*j, *j ^ *i);
  }
}
```

```cpp
// [l, r] 的區間 xor max (1-indexed)
vector<array<int, B>> bas(n + 1), lst(n + 1);
for (int i = 1; i <= n; i++) {
  int x = a[i - 1], p = i;
  bas[i] = bas[i - 1], lst[i] = lst[i - 1];
  for (int j = B - 1; j >= 0; j--)
    if (x >> j & 1) {
      if (!bas[i][j]) {
        bas[i][j] = x, lst[i][j] = p;
        break;
      }
      if (lst[i][j] < p)
        swap(lst[i][j], p), swap(x, bas[i][j]);
      x ^= bas[i][j];
    }
}
auto qry = [&](int l, int r) -> int {
  int mx = 0;
  for (int i = B - 1; i >= 0; i--)
    if (lst[r][i] >= l)
      mx = max(mx, mx ^ bas[r][i]);
  return mx;
};
```

6.8 Gaussian Elimination [5d1aa7]

```cpp
// 找反矩陣
    就開 2n，右邊放單位矩陣，做完檢查左半是不是單位，回傳右半
// 0  : no solution
// -1 : infinity solution
// 1  : one solution
template<class T> tuple<T,
    int, vector<T>> gaussianElimination(vector<vector<T>> a) {
  T det = 1;
  bool zeroDet = false;
  int n = a.size(), m = a[0].size(), rk = 0, sgn = 1;
  for (int c = 0; c < n; c++) {
    int p = -1;
    for (int r = rk; r < n; r++) {
      if (a[r][c] != 0) {
        p = r;
        break;
      }
    }
    if (p == -1) {
      zeroDet = true;
      continue;
    }
    if (p != rk) swap(a[rk], a[p]), sgn *= -1;
    det *= a[rk][c];
    T inv = 1 / a[rk][c];
    for (int j = c; j < m; j++) a[rk][j] *= inv;
    for (int r = 0; r < n; r++) {
      if (r == rk || a[r][c] == 0) continue;
      T fac = a[r][c];
      for (int j = c; j < m; j++)
        a[r][j] -= fac * a[rk][j];
    }
    rk++;
  }
  det = (zeroDet ? 0 : det * sgn);
  for (int r = rk; r < n; r++)
    if (a[r][m - 1] != 0) return {det, 0, {}};
  if (rk < n) return {det, -1, {}};
  vector<T> ans(n);
  for (int i = 0; i < n; i++) ans[i] = a[i][m - 1];
  return {det, 1, ans};
}
template<class T> tuple<int, vector
    <T>, vector<vector<T>>> findBasis(vector<vector<T>> a) {
  int n = a.size(), m = a[0].size(), rk = 0;
  vector<int> pos(m - 1, -1);
  for (int c = 0; c < m - 1; c++) {
    int p = -1;
    for (int r = rk; r < n; r++) {
      if (a[r][c] != 0) {
        p = r;
        break;
      }
    }
    if (p == -1) continue;
    if (p != rk) swap(a[rk], a[p]);
    pos[c] = rk;
    T inv = 1 / a[rk][c];
    for (int j = c; j < m; j++) a[rk][j] *= inv;
    for (int r = 0; r < n; r++) {
      if (r == rk || a[r][c] == 0) continue;
      T fac = a[r][c];
      for (int j = c; j < m; j++)
        a[r][j] -= fac * a[rk][j];
    }
    rk++;
  }
  vector<T> sol(m - 1);
  vector<vector<T>> basis;
  for (int r = rk; r < n; r++)
    if (a[r][m - 1] != 0)
      return {-1, sol, basis};
  for (int c = 0; c < m - 1; c++)
    if (pos[c] != -1)
      sol[c] = a[pos[c]][m - 1];
  for (int c = 0; c < m - 1; c++)
    if (pos[c] == -1) {
      vector<T> v(m - 1);
      v[c] = 1;
      for (int j = 0; j < m - 1; j++)
        if (pos[j] != -1)
          v[j] = -a[pos[j]][c];
      basis.push_back(v);
    }
  return {rk, sol, basis};
}
template<class T> using Matrix = vector<vector<T>>;
```

6.9 Integer Partition [83bc9d]

```cpp
// CSES_Sum_of_Divisors
const int Mod = 1E9 + 7;
const int inv_2 = 500000004;
// n / 1 * 1 + n / 2 * 2 + n / 3 * 3 + ... + n / n * n
void integerPartition() {
  ll ans = 0, n; cin >> n;
  for (ll l = 1, r; l <= n; l = r + 1) {
    r = n / (n / l);
    ll val = n / l; // n / l 到 n / r 一樣的值
    ll sum = (((l + r) %
        Mod) * ((r - l + 1) % Mod)) % Mod * inv_2; // l 加到 r
    val %= Mod; sum %= Mod;
    ans += val * sum;
    ans %= Mod;
  }
  cout << ans << "\n";
}
```

6.10 Basic Theorems

- $a^{m-1} \equiv 1 \pmod{m}$
- $a^{m-2} \equiv a^{-1} \pmod{m}$
- $\tau(N) = \prod_{i=1}^{k}(p_i^0 + p_i^1 + \cdots + p_i^{a_i})$
- $\prod\limits_{d|N} d = N^{\tau(N)}$
- Highly Composite Number:
  - $(10^6): 720720 = 2^4 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13, \tau = 240$
  - $(10^9): 735134400 = 2^6 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17, \tau = 1344$
  - $(10^{12}): 963761198400 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23, \tau = 6720$
  - $(10^{15}): 866421317361600 = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31, \tau = 26880$
  - $(10^{18}): 897612484786617600 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7^2 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37, \tau = 103680$

6.10.1 Combination

- $C(n,m) = C(n,m-1) * (n-m+1)/m$
- $C(n+1,m) = C(n,m) + C(n,m-1)$
- $C(n,k) = 1$ (mod 2) $\Leftrightarrow$ all bit of $k \leq$ all bit of n in binary
- $(x+y)^n = \sum_{k=0}^{n} C(n,k) x^{n-k} y^k$

6.10.2 Game Theorem

- sg 值為 0 代表先手必敗
- 當前 sg 值 = 經過一方操作後，另一方所有可能的後繼狀態的 mex
- 若某個後繼狀態由多個狀態構成，就 xor 起來當作此後繼狀態的 sg 值
- 單組基礎 nim 的 sg 值為本身的原因: $f(0) = 0, f(1) = mex(f(0)) = 1, f(2) = mex(f(0), f(1)) = 2...$, 都是自己
- 多組賽局可以把 sg 值 xor 起來, 當成最後的 sg 值, nim 也是一樣, 且由於 xor 性質, 如果可以快速知道 $sg(1)g(2)...g(n)$, 就可以用 xor 性質處理不連續組合

6.10.3 Pisano Period

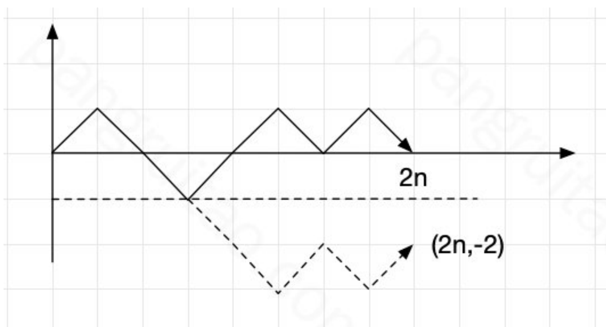
- $\pi(ab) = \text{lcm}(\pi(a), \pi(b)) \quad (\text{gcd}(a,b) = 1)$
- $\pi(p^e) \mid \pi(p) \cdot p^{e-1}$
- $\pi(p) \mid p^2 - 1 \quad (p \neq 2,5)$
- $\pi(2) = 3, \pi(5) = 20$
- so can deal with $p \approx 10^9$ in long long

6.10.4 Burnside's lemma

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

- G：各種翻轉操作所構成的置換群
- X/G：本質不同的方案的集合
- X^g：對於某一種操作 g，所有方案中，經過 g 這種翻轉後保持不變的方案的集合
- 集合取絕對值代表集合數

6.10.5 Catalan Theorem



1. n 個往上 n 個往下，先枚舉所有情況 $\frac{(2n)!}{n!n!} = C_n^{2n}$

2. 扣掉非法的，有多少種可能讓最後的點落在 $(2n,-2)$
假設往上有 x 個，往下有 y 個，會有:

$$\begin{cases} x+y=2n \\ y-x=2 \end{cases} \Rightarrow \begin{cases} x=n-1 \\ y=n+1 \end{cases}$$

所以只要扣掉 C^{2n}_{n-1} 即可

6.10.6 積性函數與反演

- 數論分塊可以快速計算一些含有除法向下取整的和式，就是像 $\sum_{i=1}^{n} f(i)g(\left\lfloor\frac{n}{i}\right\rfloor)$ 的和式。當可以在 $O(1)$ 內計算 $f(r)-f(l)$ 或已經預處理出 f 的前綴和時，數論分塊就可以在 $O(\sqrt{n})$ 的時間內計算上述和式的值。
- 迪利克雷捲積 $h(x)=\sum_{d|x} f(d)g(\frac{x}{d})$
- 積性函數
  - $\epsilon(n)$: 只在 $n=1$ 時為 1，其餘情況皆為 0。
  - 莫比烏斯函數
    1. $$\sum_{d|n}\mu(d)=\begin{cases} 1 & \text{for } n=1 \\ 0 & \text{for } n\neq 1 \end{cases}$$
    2. μ 是常數函數 1 的反元素
       $\Rightarrow \mu*1=\epsilon$，
  - ϕ 歐拉函數: x 以下與 x 互質的數量
    * $\phi(p_1^{a_1}\cdots p_k^{a_k})=(p_1^{a_1-1}(p_1-1))\cdots(p_k^{a_k-1}(p_k-1))$
    * $n=\sum_{d|n}\phi(d)$
    * $$\begin{aligned}\phi*1&=\sum_{d|n}\phi(\frac{n}{d}) \text{ 質因數分解} \\ &=\sum_{i=0}^{c}\phi(p^i) \\ &=1+p^0(p-1)+p^1(p-1)+...+p^{c-1}(p-1) \\ &=p^c \\ &=id\end{aligned}$$
- 莫比烏斯反演公式
  - $f(n)=\sum_{d|n}g(d)\Leftrightarrow g(n)=\sum_{d|n}\mu(d)f(\frac{n}{d})$
  - $f(n)=\sum_{n|d}g(d)\Leftrightarrow g(n)=\sum_{n|d}\mu(\frac{d}{n})f(d)$
- 例子

$$\begin{aligned}&\sum_{i=a}^{b}\sum_{j=c}^{d}[gcd(i,j)=k] \\ &\Rightarrow\sum_{i=1}^{x}\sum_{j=1}^{y}[gcd(i,j)=k] \\ &=\sum_{i=1}^{\left\lfloor\frac{x}{k}\right\rfloor}\sum_{j=1}^{\left\lfloor\frac{y}{k}\right\rfloor}\epsilon(gcd(i,j)) \\ &=\sum_{i=1}^{\left\lfloor\frac{x}{k}\right\rfloor}\sum_{j=1}^{\left\lfloor\frac{y}{k}\right\rfloor}\sum_{d|gcd(i,j)}\mu(d) \\ &=\sum_{d=1}^{\infty}\mu(d)\sum_{i=1}^{\left\lfloor\frac{x}{k}\right\rfloor}[d|i]\sum_{j=1}^{\left\lfloor\frac{y}{k}\right\rfloor}[d|j] \text{ d 可整除 i 時為 1} \\ &=\sum_{d=1}^{min(\left\lfloor\frac{x}{k}\right\rfloor,\left\lfloor\frac{y}{k}\right\rfloor)}\mu(d)\left\lfloor\frac{x}{kd}\right\rfloor\left\lfloor\frac{y}{kd}\right\rfloor\end{aligned}$$

6.11 Mobius Inverse [d41d8c]

```
void solve() { // pref: pref of mu
  ll a, b, c, d, k; cin >> a >> b >> c >> d >> k;
  sieve(N);
  auto cal = [&](ll x, ll y) -> int {
    int res = 0;
    for (int l = 1, r; l <= min(x, y); l = r + 1) {
      r = min(x / (x / l), y / (y / l));
      res += (pref[r] - pref[l - 1]) * (x / l) * (y / l);
    }
    return res;
  };
  cout <<
       cal(b / k, d / k) - cal((a - 1) / k, d / k) - cal(b /
      k, (c - 1) / k) + cal((a - 1) / k, (c - 1) / k) << "\n";
}
```

6.12 LCM Convolution [e11c48]

```
vector<Z> zeta(vector<Z> a) { // ak = sum_{d|k} a[d]
  int n = a.size() - 1;
  for (int p : primes)
    for (int i = 1; i <= n / p; i++) a[i * p] += a[i];
  return a;
}
vector<Z> fmt(vector<Z> a) { // ck = sum_{d|k} mu[d] * a[k/d]
  int n = a.size() - 1;
  for (int p : primes)
    for (int i = n / p; i >= 1; i--) a[i * p] -= a[i];
  return a;
}
vector<Z> conv(vector
    <Z> a, vector<Z> b) { // ck = sum_{lcm(i,j)=k} a[i] * b[j]
  int n = a.size() - 1;
  a = zeta(a), b = zeta(b);
  vector<Z> c(n + 1);
  for (int i = 1; i <= n; i++) c[i] = a[i] * b[i];
  return fmt(c);
}
```

6.13 exLucas [f8047f]

```
ll legendre(ll n, int p) { // n! 中質數 p 的冪次
  ll res = 0;
  while (n) { n /= p, res += n; }
  return res;
}
Z C_pe(ll n, ll m, int p, int e) {
  int pe = power(p, e);
  Z::setMod(pe); // inv 要用 exgcd
  vector<Z> fac(pe); fac[0] = 1;
  for (int j
      = 1; j < pe; j++) fac[j] = fac[j - 1] * (j % p ? j : 1);
  Z wilson =
       p == 2 && e >= 3 ? 1 : -1; // (p^e)! = wilson (mod p^e)
  function<Z(ll)> fastFac = [&](ll n)
  { return n ? power
      (wilson, n / pe) * fac[n % pe] * fastFac(n / p) : 1; };
  ll vp = legendre(n, p) - legendre(m, p) - legendre(n - m, p);
  if (vp >= e) return 0;
  return power
      (Z(p), vp) * fastFac(n) / (fastFac(m) * fastFac(n - m));
}
ll exlucas(ll n, ll m, int mod) {
  vector<pair<int, int>> a;
  for (int i = 2; i * i <= mod; i++) {
    if (mod % i) continue;
    int e = 0, p = 1;
    while (mod % i == 0) mod /= i, e++, p *= i;
    a.emplace_back(C_pe(n, m, i, e).x, p);
  }
  if (mod != 1) a.emplace_back(C_pe(n, m, mod, 1).x, mod);
  return CRT(a);
}
```

6.14 Quadratic Residue [da805f]

```
int jacobi(int x, int p) {
  int s = 1;
  for (; p > 1; ) {
    x %= p;
    if (x == 0) return 0;
    const int r = __builtin_ctz(x);
    if ((r & 1) && ((p + 2) & 4)) s = -s;
    x >>= r;
    if (x & p & 2) s = -s;
    swap(x, p);
  }
  return s;
}
template<class Z> int quadraticResidue(Z x) {
  int p = Z::getMod();
  if (p == 2) return x.x & 1;
  const int jc = jacobi(x.x, p);
  if (jc == 0) return 0;
  if (jc == -1) return -1;
  Z b, d;
  while (true) {
    b = rand(), d = b * b - x;
    if (jacobi(d.x, p) == -1) break;
  }
  Z f0 = b, f1 = 1, g0 = 1, g1 = 0, tmp;
  for (int e = (p + 1) >> 1; e; e >>= 1) {
    if (e & 1) {
      tmp = g0 * f0 + d * (g1 * f1);
      g1 = g0 * f1 + g1 * f0, g0 = tmp;
    }
    tmp = f0 * f0 + d * (f1 * f1);
    f1 = f0 * f1 * 2, f0 = tmp;
  }
  return min(g0.x, p - g0.x);
}
```

6.15 BSGS [ed42da]

```
// a^x = b (mod m)
// x = A * sq - B (0 <= A, B <= sq)
vector<int> BSGS(int a, int b, int m) { // gcd(a, m) = 1
  Z::setMod(m);
```

```cpp
  unordered_map<int, int> mp;
  ll sq = 1; while (sq * sq < m) sq++;
  Z rhs = b; vector<int> res;
  for (int B = 0; B <= sq; B++) {
    mp[rhs.x] = B;
    rhs *= a;
  }
  Z a_pow_sq = power(Z(a), sq), lhs = a_pow_sq;
  for (int A = 1; A <= sq; A++, lhs *= a_pow_sq)
    if (mp.find(lhs.x) != mp.end()) {
      int B = mp[lhs.x];
      res.push_back(A * sq - B);
    }
  return res;
}
int exBSGS(int a, int b, int m) {
  if (b == 1 || m == 1) return 0;
  if (!a) return b ? -1 : 1;
  int k = 0;
  ll tmp = 1;
  while (__gcd(a, m) != 1) {
    int d = __gcd(a, m);
    if (b % d != 0) return -1;
    b /= d, m /= d, k++;
    tmp = tmp * (a / d) % m;
    if (tmp == b) return k;
  }
  Z::setMod(m);
  Z b_ = Z(b) * Z(tmp).inv();
  auto xs = BSGS(a, b_.x, m);
  return xs
         .empty() ? -1 : *min_element(xs.begin(), xs.end()) + k;
}
```

7 Search and Gready

7.1 Binary Search [d41d8c]

```cpp
void binarySearch() {
  // 二分找上界
  // 如果無解會 = 原 lo, lo 要先 - 1
  while (lo < hi) {
    int x = (lo + hi + 1) / 2;
    if (check(x)) lo = x;
    else hi = x - 1;
  }
  cout << lo;
  // 二分找下界
  // 如果無解會 = 原 hi, hi 要先 + 1
  while (lo < hi) {
    int x = (lo + hi) / 2;
    if (check(m)) hi = x;
    else lo = x + 1;
  }
  cout << lo;
}
```

7.2 Ternary Search [d41d8c]

```cpp
void ternarySearch() {
  int lo = 0, hi = 10;
  while (lo < hi) {
    int xl = lo + (hi - lo) / 3;
    int xr = hi - (hi - lo) / 3;
    int resl = calc(xl), resr = calc(xr);
    if (resl < resr) {
      lo = xl + 1;
    } else {
      hi = xr - 1;
    }
  }
}
```

8 DP

8.1 Projects [154da3]

```cpp
void projects() { // 排程有權重問題，輸出價值最多且時間最少
  int n; cin >> n;
  vector<array<int, 4>> a(n + 1);
  for (int i = 1; i <= n; i++) {
    int u, v, w; cin >> u >> v >> w;
    a[i] = {v, u, w, i};
  }
  sort(a.begin(), a.end());
  vector<array<ll, 4>> dp(n + 1);
  // {w, time, 有沒選, 上個是誰}
  vector<ll> vs(n + 1);
  for (int i = 1; i <= n; i++) vs[i] = a[i][0];
  for (int i = 1; i <= n; i++) {
    int id = --lower_bound(all(vs), a[i][1]) - vs.begin();
    dp[i] = dp[i - 1];
    ll nw = dp[id][0] + a[i][2];
    ll nt = dp[id][1] + a[i][0] - a[i][1];
    if (dp[i][0] < nw || dp[i][0] == nw && dp[i][1] > nt) {
      dp[i] = {nw, nt, 1, id};
    }
  }
  vector<int> ans;
  for (int i = n; i != 0;) {
    if (dp[i][2]) {
      ans.push_back(a[i][3]);
      i = dp[i][3];
    } else i--;
  } reverse(ans.begin(), ans.end());
}
```

8.2 Bitmask [eedcfc]

```cpp
void hamiltonianPath() {
  int n, m; cin >> n >> m;
  vector<vector<int>> adj(n);
  for (int i = 0; i < m; i++) {
    int u, v; cin >> u >> v;
    adj[--v].push_back(--u);
  }
  // 以...為終點，走過...
  vector dp(n, vector<int>(1 << n));
  dp[0][1] = 1;
  for (int mask = 1; mask < 1 << n; mask++) {
    if ((mask & 1) == 0) continue;
    for (int i = 0; i < n; i++) {
      if ((mask >> i & 1) == 0) continue;
      if (i == n - 1 && mask != (1 << n) - 1) continue;
      int pre = mask ^ (1 << i);
      for (int j : adj[i]) {
        if ((pre >> j & 1) == 0) continue;
        dp[i][mask] = (dp[i][mask] + dp[j][pre]) % Mod;
      }
    }
  }
  cout << dp[n - 1][(1 << n) - 1] << "\n";
}
void minClique() { // 移掉一些邊，讓整張圖由最少團組成
  int n, m;
  cin >> n >> m;
  vector<bitset<N>> g(n);
  for (int i = 0; i < m; i++) {
    int u, v; cin >> u >> v;
    u--; v--; g[u][v] = g[v][u] = 1;
  }
  vector<int> dp(1 << n, inf);
  dp[0] = 1;
  for (int mask = 0; mask < 1 << n; mask++) { // 先正常 dp
    for (int i = 0; i < n; i++) {
      if (mask & (1 << i)) {
        int pre = mask ^ (1 << i);
        if (dp[pre] == 1 && (g[i] & bitset<N>(pre)) == pre)
          dp[mask] = 1; // i 有連到所有 pre
      }
    }
  }
  for (int mask = 0; mask < 1 << n; mask++) // 然後枚舉子集 dp
    for (int sub = mask; sub; --sub &= mask)
      dp[mask] = min(dp[mask], dp[sub] + dp[mask ^ sub]);
  cout << dp[(1 << n) - 1] << "\n";
}
```

8.3 Monotonic Queue [c9ba14]

```cpp
// 應用: dp(i) = h(i) + max(A(j)), for l(i) <= j <= r(i)
// A(j) 可能包含 dp(j), h(i) 可 O(1)
void boundedKnapsack() {
  int n, k; // O(nk)
  vector<int> w(n), v(n), num(n);
  deque<int> q;
  // 將同餘的數分在同一組
  // 每次取出連續 num[i] 格中最大值
  // g_x = max(_{k=0}^num[i] (g'_{x-k} + v_i*k))
  // G_x = g'_{x} - v_i*x
  // x 代 x-k => v_i*(x-k)
  // g_x = max(_{k=0}^num[i] (G_{x-k} + v_i*x))
  vector<vector<ll>> dp(2, vector<ll>(k + 1));
  for (int i = 0; i < n; i++) {
    for (int r = 0; r < w[i]; r++) { // 餘數
      q.clear(); // q 記錄在 x = i 時的 dp 有單調性
      for (int x = 0; x * w[i] + r <= k; x++) {
        while (!q.empty() &&
               q.front() < x - num[i]) q.pop_front(); // 維護遞減
        ll nxt = dp[0][x * w[i] + r] - x * v[i];
        while (!q.empty() && dp[0][q.back()
               * w[i] + r] - q.back() * v[i] < nxt) q.pop_back();
        q.push_back(x);
        dp[1][x * w[i] + r] = dp[0][q.
               front() * w[i] + r] - q.front() * v[i] + x * v[i];
      }
    }
    swap(dp[0], dp[1]);
  }
  cout << dp[0][k] << "\n";
}
```

8.4 Digit [1c2af1]

```cpp
// 只含 0, 1, 8 的回文數
const int N = 44;
ll memo[N + 1][N + 1]; // memset -1, once
ll solve(string s) {
  vector<int> a {0};
```

```cpp
  while (!s.empty()) {
    a.push_back(s.back() - '0');
    s.pop_back();
  }
  int len = a.size() - 1;
  vector<int> now(len + 1, -1); // 紀錄目前的數字
  return [&](this
        auto &&self, int p, int eff, bool f0, bool lim) -> ll {
    if (!p) return 1; // eff: 扣除前導 0 的長度
    if (!lim && !f0 && memo[p][eff] != -1) return memo[p][eff];
    ll res = 0;
    int lst = lim ? a[p] : 9; // or 1 for binary
    for (int i = 0; i <= lst; i++) {
      if (i != 0 && i != 1 && i != 8) continue;
      bool nlim = lim && i == lst;
      now[p] = i;
      if (f0 && i == 0) { // 處理前導零
        res += self(p - 1, eff - 1, true, nlim);
      } else if (p > eff / 2) { // 前半段
        res += self(p - 1, eff, false, nlim);
      } else if (now[p] == now[eff - p + 1]) {
        res += self(p - 1, eff, false, nlim);
      }
    }
    if (!lim && !f0) memo[p][eff] = res;
    return res;
  } (len, len, true, true);
}
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
     不前導零、枚舉 p 往後所有數字中，每個 digit x 出現的次數
ll solve(int x, ll n) {
  vector<int> a {0};
  vector<ll> sum {0}; // 貼上限時，位數 p 以下有多少數字
  while (n) {
    a.push_back(n % 10);
    sum.push_back
         (sum.back() + (n % 10) * power(10, a.size() - 2));
    n /= 10;
  }
  int len = a.size() - 1;
  return
        [&](this auto &&self, int p, bool f0, bool lim) -> ll {
    if (!p) return 0; // 貢獻來自於當前 digit
    if (!lim && !f0 && memo[p][x] != -1) return memo[p][x];
    ll res = 0;
    int lst = lim ? a[p] : 9; // or 1 for binary
    for (int i = 0; i <= lst; i++) {
      bool nlim = lim && i == lst;
      if (f0 && i == 0) { // 處理前導零
        res += self(p - 1, true, nlim);
      } else if (nlim) { // 枚舉的前幾位都貼在 n 上
        res += self(p - 1, false, nlim);
        if (i == x) res += sum[p - 1] + 1;
      } else {
        res += self(p - 1, false, nlim);
        if (i == x) res += power(10, p - 1);
      }
    }
    if (!lim && !f0) memo[p][x] = res;
    return res;
  } (len, true, true);
}
// 連續兩位 digit > 1
ll memo[N + 1][10]; // memset -1, once
// 不貼上限、
     不前導零、前一位為 prv 時、枚舉 p 往後所有數字的合法個數
ll solve(ll x) { // 建 a...
  return [&](this
        auto &&self, int p, int prv, bool f0, bool lim) -> ll {
    if (!p) return 1; // 貢獻來自於當前數字數量
    if (!lim && !f0 && memo[p][prv] != -1) return memo[p][prv];
    ll res = 0;
    int lst = lim ? a[p] : 9; // or 1 for binary
    for (int i = 0; i <= lst; i++) {
      if (prv != -1 && abs(i - prv) < 2) continue;
      bool nlim = lim && i == lst;
      if (f0 && i == 0) { // 處理前導零
        res += self(p - 1, -1, true, nlim);
      } else {
        res += self(p - 1, i, false, nlim);
      }
    }
    if (!lim && !f0) memo[p][prv] = res;
    return res;
  } (len, -1, true, true);
}
```

8.5 SOS [ec9d5a]

```cpp
// 使用情況: 跟 bit 與(被)包含有關，且 x 在 1E6 左右
// 題目: 一數組，問有多少所有數 & 起來為 0 的集合數
// dp[
     x] 代表包含 x 的 y 個數(比 x 大且 bit 1 全包含 x 的有幾個)
// 答案應該包含在 dp[0] 內，但是有重複元素，所以考慮容斥
// => ans = \sum _{i=0}^{n} (-1)^{pop_count(i)} 2^{dp[i]-1}
// => 全
     部為 0 的個數 - 至少一個為 1 的個數 + 至少兩個為 1 的個數
void solve(int n, vector<int> a) {
  int m = __lg(*max_element(a.begin(), a.end())) + 1;
  // 定義 dp[mask] 為 mask 被包含於 a[i] 的 i 個數
  vector<Z> dp(1 << m);
  for (int i = 0; i < n; i++) dp[a[i]] += 1;
  for (int i = 0; i < m; i++) {
    for (int mask = 0; mask < 1 << m; mask++) {
      if (mask >> i & 1) {
        int pre = mask ^ (1 << i);
        dp[pre] += dp[mask];
      }
    }
  }
  for (int i = 0; i < 1 << m; i++) {
    dp[i] = power(Z(2), int(dp[i])) - 1;
  }
  // 排容，dp[i]: & 起來為 i 的集合數
  for (int i = 0; i < m; i++) {
    for (int mask = 0; mask < 1 << m; mask++) {
      if (mask >> i & 1) {
        int pre = mask ^ (1 << i);
        dp[pre] -= dp[mask];
      }
    }
  }
}
// x | y = x, 代表包含於 x 的 y 個數, 定義為 dp[x][0]
// x & y = x, 代表包含 x 的 y 個數, 定義為 dp[x][1]
// x & y
      != 0, 代表至少有一個位元都為 1 的 y 個數, = n - dp[~x][0]
void solve(int n, vector<int> a) {
  int m = __lg(*max_element(a.begin(), a.end())) + 1;
  vector<array<ll, 2>> dp(1 << m);
  for (int i = 0; i < n; i++) dp[a[i]][0]++, dp[a[i]][1]++;
  for (int i = 0; i < m; i++) {
    for (int mask = 0; mask < 1 << m; mask++) {
      if (mask >> i & 1) {
        int pre = mask ^ (1 << i);
        dp[mask][0] += dp[pre][0];
        dp[pre][1] += dp[mask][1];
      }
    }
  }
}
```

8.6 CHT [bdb950]

```cpp
// 應用: dp(x) = C(x) + min/max(A(i) * x + B(i)), for i < x
// 斜率遞減，求 min，上凸包，自行轉化問題
// ex: m 遞增取 max => max(mx + b) = -min(-m x - b)
struct Line { // x 盡量從 1 開始
  ll m, b;
  Line(ll m = 0, ll b = 0) : m(m), b(b) {}
  ll eval(ll x) { return m * x + b; }
};
struct CHT { // 斜率單調
  deque<Line> hull;
  CHT() {}
  CHT(Line init) { addLine(init); }
  bool frontBad(Line &l1, Line &l2, ll x) {
    return l1.eval(x) >= l2.eval(x);
  }
  bool backBad(Line &l1, Line &l2, Line &l) {
    // 只要
         l2 跟 l 的 x 交點 <= l1 跟 l 的 x 交點，l2 就用不到了
    return (l.b
         - l2.b) * (l1.m - l.m) <= (l.b - l1.b) * (l2.m - l.m);
  }
  void addLine(Line l) {
    if (hull.size() > 0 && hull.back().m == l.m) {
      if (hull.back().b <= l.b) return;
      else hull.pop_back();
    }
    while (hull.size() > 1 && backBad(hull
         [hull.size() - 2], hull.back(), l)) hull.pop_back();
    hull.push_back(l);
  }
  ll query(ll x) { // 查詢沒單調性需要二分搜
    while (hull.size() >
         0 && frontBad(hull[0], hull[1], x)) hull.pop_front();
    return hull[0].eval(x);
  }
};
```

8.7 DNC [a29240]

```cpp
// 應用: 切 k 段問題，且滿足四邊形不等式
// w(a,c) + w(b,d) <= w(a,d) + w(b,c)
// dp[k][j] = min(dp[k - 1][i] + cost[i][j])
// O(K N logN)
auto getCost = [&](int l, int r) {}; // (l, r], 1-based
vector<ll> dp(n + 1, inf); dp[0] = 0;
for (int j = 0; j < k; j++) {
  vector<ll> ndp(n + 1, inf);
  [&](this auto &&self, int l, int r, int optl, int optr) {
    if (l > r) return;
    int m = (l + r) / 2, opt = 0;
    ndp[m] = inf;
    for (int i = optl; i <= min(m - 1, optr); i++) {
```

```cpp
      ll x = dp[i] + getCost(i, m);
      if (x < ndp[m]) ndp[m] = x, opt = i;
    }
    self(l, m - 1, optl, opt);
    self(m + 1, r, opt, optr);
  } (1, n, j, n - 1);
  dp = ndp;
}
```

8.8 LiChao Segment Tree [2a9325]

```cpp
// 應用: dp(i) = h(i) + min/max(A(j)X(i) + B(j)), for j <= r(i)
//         y    =  c   +         m   x    +   b
template<class T, class F = less<ll>>
struct LiChaoSeg {
  F cmp = F();
  static const T inf = max(numeric_limits
      <T>::lowest() / 2, numeric_limits<T>::max() / 2, F());
  struct Line {
    T m, b;
    Line(T m = 0, T b = inf) : m(m), b(b) {}
    T eval(T x) const { return m * x + b; }
  };
  struct Node {
    Line line;
    ll l = -1, r = -1;
  };
  ll n;
  vector<Node> nd;
  LiChaoSeg(ll n) : n(n) { newNode(); }
  void addLine(Line line) { update(0, 0, n, line); }
  void rangeAddLine(Line line
    , ll ql, ll qr) { rangeUpdate(0, 0, n, ql, qr, line); }
  T query(ll x) { return query(x, 0, 0, n); }
private:
  int newNode() {
    nd.emplace_back();
    return nd.size() - 1;
  }
  void update(int p, ll l, ll r, Line line) {
    ll m = (l + r) / 2;
    bool left = cmp(line.eval(l), nd[p].line.eval(l));
    bool mid = cmp(line.eval(m), nd[p].line.eval(m));
    if (mid) swap(nd[p].line, line);
    if (r - l == 1) return;
    if (left != mid) {
      if (nd[p].l == -1) nd[p].l = newNode();
      update(nd[p].l, l, m, line);
    } else {
      if (nd[p].r == -1) nd[p].r = newNode();
      update(nd[p].r, m, r, line);
    }
  }
  void rangeUpdate
      (int p, ll l, ll r, ll ql, ll qr, Line line) {
    if (r <= ql || l >= qr) return;
    if (ql <= l && r <= qr) return update(p, l, r, line);
    if (nd[p].l == -1) nd[p].l = newNode();
    if (nd[p].r == -1) nd[p].r = newNode();
    ll m = (l + r) / 2;
    rangeUpdate(nd[p].l, l, m, ql, qr, line);
    rangeUpdate(nd[p].r, m, r, ql, qr, line);
  }
  T query(ll x, int p, ll l, ll r) {
    if (p == -1) return inf;
    ll m = (l + r) / 2;
    if (x < m) return
        min(nd[p].line.eval(x), query(x, nd[p].l, l, m), cmp);
    else return
        min(nd[p].line.eval(x), query(x, nd[p].r, m, r), cmp);
  }
};
```

9 Geometry

9.1 Basic [0f20c9]

```cpp
const double eps = 1E-9;
template<class T> struct Pt {
  T x, y;
  Pt(T x = 0, T y = 0) : x(x), y(y) {}
  Pt operator-() const { return Pt(-x, -y); }
  Pt operator+(Pt p) const { return Pt(x + p.x, y + p.y); }
  Pt operator-(Pt p) const { return Pt(x - p.x, y - p.y); }
  Pt operator*(T k) const { return Pt(x * k, y * k); }
  Pt operator/(T k) const { return Pt(x / k, y / k); }
  bool operator==(Pt p) const { return x == p.x && y == p.y; }
  bool operator!=(Pt p) const { return x != p.x || y != p.y; }
  friend istream &operator>>(istream &is, Pt &p)
  { return is >> p.x >> p.y; }
  friend ostream &operator<<(ostream &os, const Pt &p)
  { return os << "(" << p.x << ", " << p.y << ")"; }
};
int sign(double x)
{ return fabs(x) <= eps ? 0 : (x > 0 ? 1 : -1); }
using P = Pt<double>;

struct Line { P a, b; };
template<class T>
     T dot(Pt<T> a, Pt<T> b) { return a.x * b.x + a.y * b.y; }
template<class T> T
     cross(Pt<T> a, Pt<T> b) { return a.x * b.y - a.y * b.x; }
template<class T> T abs2(Pt<T> p) { return dot(p, p); }
double abs(P p) { return sqrt(abs2(p)); }
double dist(P a, P b) { return abs(a - b); }
double abs(Line l) { return abs(l.a - l.b); }
int dir(P p, Line l) // left -1, right 1, on 0
{ return -sign(cross(l.b - l.a, p - l.a)); }
bool btw(P p, Line l) // c on segment ab?
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P norm(P p) { return p / abs(p); }
P rot(P p) { return { -p.y, p.x }; } // 90 degree CCW
P rot(P p, double d) {
  double c = cos(d), s = sin(d);
  return P(p.x * c - p.y * s, p.x * s + p.y * c);
}

bool parallel(Line l1, Line l2)
{ return sign(cross(l1.b - l1.a, l2.b - l2.a)) == 0; }
P lineIntersection(Line l1, Line l2)
{ return l1.a + (l1.b - l1.a) * (cross(l2.b
     - l2.a, l1.a - l2.a) / cross(l2.b - l2.a, l1.a - l1.b)); }
bool pointOnSegment(P p, Line l)
{ return dir(p, l) == 0 && sign(dot(p - l.a, p - l.b)) <= 0; }
P projvec(P p, Line l) {
  P v = l.b - l.a;
  return l.a + v * (dot(p - l.a, v) / abs2(v));
}
// 0 : not intersect
// 1 : strictly intersect
// 2 : overlap
// 3 : intersect at endpoint
tuple<int, P, P> segmentIntersection(Line l1, Line l2) {
  if (sign(max(l1.a.x, l1.b.x) - min(l2.a.x, l2.b.x)) == -1 ||
    sign(min(l1.a.x, l1.b.x) - max(l2.a.x, l2.b.x)) == 1 ||
    sign(max(l1.a.y, l1.b.y) - min(l2.a.y, l2.b.y)) == -1 ||
    sign(min(l1.a.y, l1.b.y) - max(l2.a.y, l2.b.y)) == 1)
    return {0, {}, {}};
  if (parallel(l1, l2)) {
    if (dir(l1.a, l2) != 0) return {0, {}, {}};
    else {
      vector<P> res;
      if (pointOnSegment(l1.a, l2)) res.push_back(l1.a);
      if (pointOnSegment(l1.b, l2)) res.push_back(l1.b);
      if (pointOnSegment(l2.a, l1)) res.push_back(l2.a);
      if (pointOnSegment(l2.b, l1)) res.push_back(l2.b);
      if (res.empty()) return {0, {}, {}};
      sort(res.begin(), res.end(), [](const P &a, const P &b)
      { return
           sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
      res.resize(unique(res.begin(), res.end()) - res.begin());
      if (res.size() == 1) return {3, res[0], res[0]};
      else return {2, res.front(), res.back()};
    }
  }
  auto cp1 = dir(l2.b, l1), cp2 =
       dir(l2.a, l1), cp3 = dir(l1.b, l2), cp4 = dir(l1.a, l2);
  if ((cp1 > 0 && cp2 > 0) ||
    (cp1 < 0 && cp2 < 0) ||
    (cp3 > 0 && cp4 > 0) ||
    (cp3 < 0 && cp4 < 0)) return {0, {}, {}};
  P p = lineIntersection(l1, l2);
  if (cp1 !=
       0 && cp2 != 0 && cp3 != 0 && cp4 != 0) return {1, p, p};
  else return {3, p, p};
}

vector<P> convexHull(vector<P> a) {
  sort(a.begin(), a.end(), [](const P &a, const P &b)
  { return sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
  a.resize(unique(a.begin(), a.end()) - a.begin());
  if (a.size() <= 1) return a;
  vector<P> h(a.size() * 2);
  int s = 0, t = 0;
  for (int i = 0; i < 2; i++, s = --t) {
    for (P p : a) {
      while (t >= s + 2 && sign(
           cross(h[t - 1] - h[t - 2], p - h[t - 2])) <= 0) t--;
      h[t++] = p;
    }
    reverse(a.begin(), a.end());
  }
  return {h.begin(), h.begin() + t};
}

double distPL(P p, Line l)
{ return abs(cross(l.a - l.b, l.a - p)) / abs(l); }
double distancePS(P p, Line l) {
  if (sign(dot(p - l.a, l.b - l.a)) < 0) return dist(p, l.a);
  if (sign(dot(p - l.b, l.a - l.b)) < 0) return dist(p, l.b);
  return distPL(p, l);
}
double distanceSS(Line l1, Line l2) {
  if (get<0>(segmentIntersection(l1, l2)) != 0) return 0.0;
  return min({distancePS(l1.a, l2), distancePS(
       l1.b, l2), distancePS(l2.a, l1), distancePS(l2.b, l1)});
}

bool lineIntersectsPolygon(Line l, const vector<P> &p) {
  int n = p.size();
  for (int i = 0; i < n; i++) {
    P a = p[i], b = p[(i + 1) % n];
    if (dir(a, l) == 0 || dir(b, l) == 0) return true;
    if ((dir(a, l) < 0) ^ (dir(b, l) < 0)) return true;
```

```
  }
  return false;
}
bool pointInPolygon(P a, const vector<P> &p) {
  int n = p.size(), t = 0;
  for (int i = 0; i < n; i++)
    if (pointOnSegment(a, {p[i], p[(i + 1) % n]})) return true;
  for (int i = 0; i < n; i++) {
    P u = p[i], v = p[(i + 1) % n];
    if (u.x < a.x && v.x >= a.x && dir(a, {v, u}) < 0) t ^= 1;
    if (u.x >= a.x && v.x < a.x && dir(a, {u, v}) < 0) t ^= 1;
  }
  return t == 1;
}
// 0 : strictly outside
// 1 : on boundary
// 2 : strictly inside
int pointInConvexPolygon(P a, const vector<P> &p) {
  int n = p.size();
  if (n == 0) return 0;
  else if (n <= 2) return pointOnSegment(a, {p[0], p.back()});
  if (pointOnSegment(a, {p[0],
       p[1]}) || pointOnSegment(a, {p[0], p[n - 1]})) return 1;
  else if (dir(a, {p[0],
        p[1]}) < 0 || dir(a, {p[0], p[n - 1]}) < 0) return 0;
  int lo = 1, hi = n - 2;
  while (lo < hi) {
    int x = (lo + hi + 1) / 2;
    if (dir(a, {p[0], p[x]}) < 0) lo = x;
    else hi = x - 1;
  }
  if (dir(a, {p[lo], p[lo + 1]}) < 0) return 2;
  else return pointOnSegment(a, {p[lo], p[lo + 1]});
}
bool segmentInPolygon(Line l, const vector<P> &p) {
  int n = p.size();
  vector<P> a {l.a, l.b};
  for (int i = 0; i < n; i++) {
    auto [t, p1,
          p2] = segmentIntersection(l, {p[i], p[(i + 1) % n]});
    if (t == 1 || t == 3) a.push_back(p1);
    else if (t == 2) a.push_back(p1), a.push_back(p2);
  }
  sort(a.begin(), a.end(), [](const P &a, const P &b)
  { return sign(a.x - b.x) == 0 ? a.y < b.y : a.x < b.x; });
  for (int i = 0; i + 1 < a.size(); i++) {
    if (sign(dist(a[i], a[i + 1])) == 0) continue;
    if (!pointInPolygon
        ((a[i] + a[i + 1]) / 2, p)) return false;
  }
  return true;
}
// non-strict
      -> >= 都改 >, assert 拿掉, 在外面特判直線(area == 0)
vector<P> hp(vector<Line> lines) {
  auto sgn = [](P p
       ) { return p.y > 0 || (p.y == 0 && p.x > 0) ? 1 : -1; };
  sort(lines.begin(), lines.end(), [&](auto l1, auto l2) {
    auto d1 = l1.b - l1.a;
    auto d2 = l2.b - l2.a;
    if (sgn(d1) != sgn(d2)) return sgn(d1) == 1;
    return cross(d1, d2) > 0;
  });
  deque<Line> ls;
  deque<P> ps;
  for (auto l : lines) {
    if (ls.empty()) {
      ls.push_back(l);
      continue;
    }
    while (!ps.empty() &&
          dir(ps.back(), l) >= 0) ps.pop_back(), ls.pop_back();
    while (!ps.empty()
         && dir(ps[0], l) >= 0) ps.pop_front(), ls.pop_front();
    if (sign(cross(
         l.b - l.a, ls.back().b - ls.back().a)) == 0) { // 平行
      if (sign(dot(l
           .b - l.a, ls.back().b - ls.back().a)) > 0) { // 同向
        if (dir(ls.back().a, l) >= 0) { // l 在內側
          assert(ls.size() == 1);
          ls[0] = l;
        }
        continue;
      }
      return {};
    }
    ps.push_back(lineIntersection(ls.back(), l));
    ls.push_back(l);
  }
  while (!ps.empty() && dir(ps.back(), ls[0]) >= 0)
    ps.pop_back(), ls.pop_back();
  if (ls.size() <= 2) return {};
  ps.push_back(lineIntersection(ls[0], ls.back()));
  return vector(ps.begin(), ps.end());
}
```

9.2 Min Euclidean Distance [2e8f35]

```
void solve(int n, vector<P> &a) { // recursive solution
  sort(a.begin(), a.end(), [](const P &a, const P &b)
  { return a.x == b.x ? a.y < b.y : a.x < b.x; });
  vector<P> t(n);
  cout << [&](this auto &&self, int l, int r) -> ll {
    if (l == r) return 9E18;
    int p = 0, m = (l + r) / 2;
    ll mid = a[m].x;
    ll ans = min(self(l, m), self(m + 1, r));
    merge(a.begin() + l, a.begin() + m + 1,
      a.begin() + m + 1, a.begin() + r + 1,
      t.begin()
           , [](const P &a, const P &b) { return a.y < b.y; });
    copy(t.begin(), t.begin() + r - l + 1, a.begin() + l);
    for (int i = l; i <= r; i++)
      if ((mid
           - a[i].x) * (mid - a[i].x) <= ans) t[p++] = a[i];
    for (int i = 0; i < p; i++) {
      for (int j = i + 1; j < p; j++) {
        if ((t[i].y - t[j].y) * (t[i].y - t[j].y) > ans) break;
        ans = min(ans, abs2(t[i] - t[j]));
      }
    }
    return ans;
  } (0, n - 1) << "\n";
}
// K-D tree solution
struct Info {
  static constexpr int DIM = 2;
  array<ll, DIM> x, L, R;
  ll distl, distr;
  ll f(const Info &i) {
    ll ret = 0;
    if (i.
         L[0] > x[0]) ret += (i.L[0] - x[0]) * (i.L[0] - x[0]);
    if (i.
         R[0] < x[0]) ret += (x[0] - i.R[0]) * (x[0] - i.R[0]);
    if (i.
         L[1] > x[1]) ret += (i.L[1] - x[1]) * (i.L[1] - x[1]);
    if (i.
         R[1] < x[1]) ret += (x[1] - i.R[1]) * (x[1] - i.R[1]);
    return ret;
  }
  void pull(const Info &l, const Info &r) {
    distl = f(l), distr = f(r);
  }
};
struct KDTree { // 1-indexed
  static constexpr int DIM = Info::DIM;
  int n, rt;
  vector<Info> info;
  vector<int> l, r;
  KDTree(const vector<Info> &
       info) : n(info.size()), info(info), l(n + 1), r(n + 1) {
    rt = build(1, n);
  }
  void pull(int p) {
    info[p].L = info[p].R = info[p].x;
    info[p].pull(info[l[p]], info[r[p]]);
    for (int ch : {l[p], r[p]}) {
      if (!ch) continue;
      for (int k = 0; k < DIM; k++) {
        info[p].L[k] = min(info[p].L[k], info[ch].L[k]);
        info[p].R[k] = max(info[p].R[k], info[ch].R[k]);
      }
    }
  }
  int build(int l, int r) {
    if (r == l) return 0;
    int m = (l + r) / 2;
    array<double, DIM> av = {}, va = {};
    for (int i = l; i < r; i++)
      for (int d = 0; d < DIM; d++)
        av[d] += info[i].x[d];
    for (int d = 0; d < DIM; d++)
      av[d] /= (double)(r - l);
    for (int i = l; i < r; i++)
      for (int d = 0; d < DIM; d++)
        va[d] +=
             (info[i].x[d] - av[d]) * (info[i].x[d] - av[d]);
    int dep = max_element(va.begin(), va.end()) - va.begin();
    nth_element
         (info.begin() + l, info.begin() + m, info.begin() + r,
      [&](const Info
           &x, const Info &y) { return x.x[dep] < y.x[dep]; });
    this->l[m] = build(l, m);
    this->r[m] = build(m + 1, r);
    pull(m); return m;
  }
  ll ans = 9E18;
  ll dist(int a, int b) {
    return (info[a].
         x[0] - info[b].x[0]) * (info[a].x[0] - info[b].x[0]) +
    (info[a].
         x[1] - info[b].x[1]) * (info[a].x[1] - info[b].x[1]);
  }
  void query(int p, int x) {
    if (!p) return;
    if (p != x) ans = min(ans, dist(x, p));
    ll distl = info[x].f(info[l[p]]);
    ll distr = info[x].f(info[r[p]]);
    if (distl < ans && distr < ans) {
      if (distl < distr) {
        query(l[p], x);
        if (distr < ans) query(r[p], x);
```

```
      } else {
        query(r[p], x);
        if (distl < ans) query(l[p], x);
      }
    } else {
      if (distl < ans) query(l[p], x);
      if (distr < ans) query(r[p], x);
    }
  }
};
```

9.3 Max Euclidean Distance [7e3b52]

```
tuple<ll, int, int> maxEuclideanDistance(vector<P> a) {
  auto get = [&](P p, Line l) -> ll
  { return abs(cross(l.a - l.b, l.a - p)); };
  ll res = 0; int n = a.size(), x, y, id = 2;
  a.push_back(a.front());
  if (n == 1) return {0, 0, 0};
  if (n == 2) return {abs2(a[0] - a[1]), 0, 1};
  for (int i = 0; i < n; i++) {
    while (get(a[id], {a[i],
          a[i + 1]}) <= get(a[(id + 1) % n], {a[i], a[i + 1]}))
      id = (id + 1) % n;
    if (res < abs2(a[i] - a[id])) {
      res = abs2(a[i] - a[id]);
      x = i, y = id;
    }
    if (res < abs2(a[i + 1] - a[id])) {
      res = abs2(a[i + 1] - a[id]);
      x = i + 1, y = id;
    }
  }
  return {res, x % n, y};
}
```

9.4 Min Circle Cover [96a4b7]

```
pair<double, P> minCircleCover(int n, vector<P> a) {
    shuffle(a.begin(), a.end(), rng);
    P c = a[0]; double r = 0;
    for (int
          i = 1; i < n; i++) if (sign(abs(c - a[i]) - r) > 0) {
    c = a[i], r = 0;
    for (int
          j = 0; j < i; j++) if (sign(abs(c - a[j]) - r) > 0) {
      c = (a[i] + a[j]) / 2.0;
      r = abs(c - a[i]);
      for (int k
            = 0; k < j; k++) if (sign(abs(c - a[k]) - r) > 0) {
        P p = (a[j] + a[i]) / 2;
        P q = (a[j] + a[k]) / 2;
        if (parallel({a[i], a[j]}, {a[j], a[k]})) continue;
        c = lineIntersection({p,
              p + rot(a[j] - a[i])}, {q, q + rot(a[k] - a[j])});
        r = abs(c - a[i]);
      }
    }
  }
    return {r, c};
}
```

9.5 Min Rectangle Cover [3a0f29]

```
pair<double, vector<P>> minRectangleCover(vector<P> p) {
  if (p.size() <= 2) return {0, {}};
  auto get = [&](P p, Line l) -> double {
    return abs(cross(l.a - l.b, l.a - p));
  }; // line 到 p 圍成的四邊形面積
  int n = p.size(), j = 2, l = 1, r = 1;
  p.push_back(p.front());
  double ans = 8E18;
  vector<P> ps;
  for (int i = 0; i < n; i++) {
    while (get(p[j], {p[i],
          p[i + 1]}) <= get(p[(j + 1) % n], {p[i], p[i + 1]}))
      j = (j + 1) % n;
    while (dot(p[i + 1] - p[i], p[r] -
          p[i]) <= dot(p[i + 1] - p[i], p[(r + 1) % n] - p[i]))
      r = (r + 1) % n;
    if (i == 0) l = j;
    while (dot(p[i + 1] - p[i], p[l] -
          p[i]) >= dot(p[i + 1] - p[i], p[(l + 1) % n] - p[i]))
      l = (l + 1) % n;
    double area = get(p[j], {p[i], p[i + 1]});
    double w = dot(p[i] - p[i +
          1], p[l] - p[i]) + dot(p[i + 1] - p[i], p[r] - p[i]);
    area *= w / abs2(p[i + 1] - p[i]);
    if (area < ans) {
      ps.clear(), ans = area;
      Line l1 {p[i], p[i + 1]};
      for (auto u : {p[r], p[j], p[l], p[i]}) {
        if (u == l1.b) {
          ps.push_back(u);
          l1 = {u, u + rot(l1.b - l1.a)};
        } else {
          Line l2 = {u, u + rot(l1.b - l1.a)};
          P res = lineIntersection(l1, l2);
          ps.push_back(res);
          l1 = {res, u};
        }
      }
    }
  }
  return {ans, ps};
}
```

9.6 Convex Polygon Distance [92874d]

```
// CCW needed, O(n + m)
double convexPolyDist(vector<P> a, vector<P> b) {
    int n = a.size(), m = b.size();
    int sa = 0, sb = 0;
    for (int i = 1; i < n; i++) if (a[i].y < a[sa].y) sa = i;
    for (int i = 1; i < m; i++) if (b[i].y > b[sb].y) sb = i;
    double ans = 1E18;
    for (int i = 0; i < n; i++) {
        P na = a[sa], nb = a[(sa + 1) % n];
        P eA = nb - na;
        while (sign(cross(eA, b[(sb + 1) % m] - b[sb])) > 0)
            sb = (sb + 1) % m;
        chmin(ans
            , distanceSS({na, nb}, {b[sb], b[(sb + 1) % m]}));
        sa = (sa + 1) % n;
    }
    return ans;
}
```

9.7 Lattice Points [fbebcc]

```
// (interior, boundary)
pair<ll, ll> latticePoints(int n, vector<P> p, ll area2) {
    ll bnd = 0;
    p.push_back(p[0]);
    for (int i = 0; i < n; i++)
        bnd += gcd(abs
            (p[i + 1].x - p[i].x), abs(p[i + 1].y - p[i].y));
    return {(area2 - bnd) / 2 + 1, bnd};
}
```

9.8 Polygon Union Area [24443d]

```
double polygonUnion(vector<vector<P>> ps) { // CCW needed
  int n = ps.size();
  for (auto &v : ps) v.push_back(v[0]);
  double res = 0;
  auto seg = [&](P o, P a, P b) -> double {
    if (b.x - a.x == 0) return (o.y - a.y) / (b.y - a.y);
    return (o.x - a.x) / (b.x - a.x);
  };
  for (int pi = 0; pi < n; pi++) {
    for (int i = 0; i + 1 < ps[pi].size(); i++) {
      vector<pair<double, int>> e {{0, 0}, {1, 0}};
      for (int pj = 0; pj < n; pj++) {
        if (pi == pj) continue;
        for (int j = 0; j + 1 < ps[pj].size(); j++) {
          auto c1 = cross(ps
               [pi][i + 1] - ps[pi][i], ps[pj][j] - ps[pi][i]);
          auto c2 = cross(ps[pi][
               i + 1] - ps[pi][i], ps[pj][j + 1] - ps[pi][i]);
          if (c1 == 0 && c2 == 0) {
            if (dot(ps[pi][i + 1] - ps[pi][i], ps[pj
                 ][j + 1] - ps[pj][j]) > 0 && (pi - pj) > 0) {
              e.emplace_back(seg
                   (ps[pj][j], ps[pi][i], ps[pi][i + 1]), 1);
              e.emplace_back(seg(ps
                   [pj][j + 1], ps[pi][i], ps[pi][i + 1]), -1);
            }
          } else {
            auto s1 = cross(ps[pj
                 ][j + 1] - ps[pj][j], ps[pi][i] - ps[pj][j]);
            auto s2 = cross(ps[pj][j
                  + 1] - ps[pj][j], ps[pi][i + 1] - ps[pj][j]);
            if (c1 >= 0
                 && c2 < 0) e.emplace_back(s1 / (s1 - s2), 1);
            else if (c1 < 0 &&
                 c2 >= 0) e.emplace_back(s1 / (s1 - s2), -1);
          }
        }
      }
      sort(e.begin(), e.end());
      double pre = clamp(e[0].first, 0.0, 1.0), sum = 0;
      int cov = e[0].second;
      for (int j = 1; j < e.size(); j++) {
        double now = clamp(e[j].first, 0.0, 1.0);
        if (!cov) sum += now - pre;
        cov += e[j].second;
        pre = now;
      }
      res += cross(ps[pi][i], ps[pi][i + 1]) * sum;
    }
  }
  return res / 2;
}
```

10 Polynomial

10.1 FFT [8d8ca2]

```
const double PI = acos(-1.0);
using cd = complex<double>;
vector<int> rev;
void fft(vector<cd> &a, bool inv = false) {
```

```
  int n = a.size();
  if (int(rev.size()) != n) {
    int k = __builtin_ctz(n) - 1;
    rev.resize(n);
    for (int i = 0; i < n; i++)
      rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
  }
  for (int i = 0; i < n; i++)
    if (rev[i] < i) swap(a[i], a[rev[i]]);
  for (int k = 1; k < n; k *= 2) {
    double ang = (inv ? -1 : 1) * PI / k;
    cd wn(cos(ang), sin(ang));
    for (int i = 0; i < n; i += 2 * k) {
      cd w(1);
      for (int j = 0; j < k; j++, w = w * wn) {
        cd u = a[i + j];
        cd v = a[i + j + k] * w;
        a[i + j] = u + v;
        a[i + j + k] = u - v;
      }
    }
  }
  if (inv) for (auto &x : a) x /= n;
}
// i + j = k, res[1] = a[1] * b[0] + a[0] * b[1]
// reverse b for i - j = k, res[k + b.size() - 1]
template<class T>
vector<T> Multiple(const vector<T> &a, const vector<T> &b) {
  vector<cd> fa(a.begin(), a.end()), fb(b.begin(), b.end());
  int n = 1, tot = a.size() + b.size() - 1;
  while (n < tot) n *= 2;
  fa.resize(n), fb.resize(n);
  fft(fa), fft(fb);
  for (int i = 0; i < n; i++)
    fa[i] = fa[i] * fb[i];
  fft(fa, true);
  vector<T> res(tot);
  for (int i = 0; i < tot; i++)
    res[i] = fa[i].real(); // use llround if need
  return res;
}
```

10.2 NTT [535f81]

```
template<int P = 998244353, int G = 3>
struct Poly : public vector<Mint<P>> {
  using Z = Mint<P>;
  static vector<int> rev;
  static vector<Z> w;
  static void ntt(vector<Z> &a, bool inv = false) {
    int n = a.size();
    if (rev.size() != n) {
      int k = __builtin_ctz(n) - 1;
      rev.resize(n);
      for (int i = 0; i
            < n; i++) rev[i] = rev[i >> 1] >> 1 | (i & 1) << k;
    }
    for (int i =
        0; i < n; i++) if (rev[i] < i) swap(a[i], a[rev[i]]);
    if (w.size() < n) {
      int k = __builtin_ctz(w.size());
      w.resize(n);
      while ((1 << k) < n) {
        Z u = power(Z(G), (P - 1) >> (k + 1));
        for (int i = 1 << (k - 1); i < (1 << k); i++) {
          w[i * 2] = w[i];
          w[i * 2 + 1] = w[i] * u;
        }
        k++;
      }
    }
    for (int k = 1; k < n; k *= 2) {
      for (int i = 0; i < n; i += 2 * k) {
        for (int j = 0; j < k; j++) {
          Z u = a[i + j], v = a[i + j + k] * w[k + j];
          a[i + j] = u + v; a[i + j + k] = u - v;
        }
      }
    }
    if (inv) {
      reverse(a.begin() + 1, a.end());
      Z inv_n = Z(n).inv();
      for (auto &x : a) x *= inv_n;
    }
  }
  explicit Poly(int n = 0) : vector<Z>(n) {}
  Poly(const vector<Z> &a) : vector<Z>(a) {}
  Poly(const initializer_list<Z> &a) : vector<Z>(a) {}
  template<class InputIt, class = _RequireInputIter<InputIt>>
  Poly(InputIt first, InputIt last) : vector<Z>(first, last) {}
  Poly operator-() const {
    vector<Z> res(this->size());
    for (int
          i = 0; i < int(res.size()); i++) res[i] = -(*this)[i];
    return Poly(res);
  }
  friend Poly operator+(Poly a, Poly b) {
    a.resize(max(a.size(), b.size()));
    for (int i = 0; i < b.size(); i++) a[i] += b[i];
    return a;
  }
  friend Poly operator-(Poly a, Poly b) {
    a.resize(max(a.size(), b.size()));
    for (int i = 0; i < b.size(); i++) a[i] -= b[i];
    return a;
  }
  friend Poly operator*(Poly a, Poly b) {
    if (a.empty() || b.empty()) return Poly();
    if (a.size() < b.size()) swap(a, b);
    int n = 1, tot = a.size() + b.size() - 1;
    while (n < tot) n *= 2;
    a.resize(n), b.resize(n);
    ntt(a), ntt(b);
    for (int i = 0; i < n; i++) a[i] *= b[i];
    ntt(a, true);
    a.resize(tot);
    return a;
  }
  friend Poly operator*(Poly a, Z x) {
    for (int i = 0; i < int(a.size()); i++) a[i] *= x;
    return a;
  }
  friend Poly operator/(Poly a, Z x) {
    for (int i = 0; i < int(a.size()); i++) a[i] /= x;
    return a;
  }
  Poly &operator+=(Poly a)
  { return (*this) = (*this) + a; }
  Poly &operator-=(Poly a)
  { return (*this) = (*this) - a; }
  Poly &operator*=(Poly a)
  { return (*this) = (*this) * a; }
  Poly &operator*=(Z a)
  { return (*this) = (*this) * a; }
  Poly &operator/=(Z a)
  { return (*this) = (*this) / a; }
  Poly shift(int k) const {
    if (k >= 0) {
      auto b = *this;
      b.insert(b.begin(), k, 0);
      return b;
    } else if (this->size() <= -k) {
      return Poly();
    } else {
      return Poly(this->begin() + (-k), this->end());
    }
  }
  Poly trunc(int k) const {
    Poly f = *this; f.resize(k); return f;
  }
  Poly deriv() const {
    if (this->empty()) return Poly();
    Poly res(this->size() - 1);
    for (int i = 0; i < this->size() - 1; i++)
      res[i] = (*this)[i + 1] * (i + 1);
    return res;
  }
  Poly integr() const {
    Poly res(this->size() + 1);
    for (int i = 0; i < this->size(); i++)
      res[i + 1] = (*this)[i] / (i + 1);
    return res;
  }
  Poly inv(int m) const {
    Poly x{(*this)[0].inv()};
    int k = 1;
    while (k < m) {
      k *= 2;
      x = (x * (Poly{2} - trunc(k) * x)).trunc(k);
    }
    return x.trunc(m);
  }
  Poly log(int m) const {
    return (deriv() * inv(m)).integr().trunc(m);
  }
  Poly pow(ll k, int m) const {
    if (k == 0) { Poly res(m); res[0] = 1; return res; }
    int i = 0;
    while (i < this->size() && (*this)[i] == 0) i++;
    if (i == this
          ->size() || i > 0 && k > (m - 1) / i) return Poly(m);
    Z v = (*this)[i];
    auto f = shift(-i) * v.inv();
    return (f.log(m - i *
            k) * Z(k)).exp(m - i * k).shift(i * k) * power(v, k);
  }
  Poly sqrt(int m) const { // need quadraticResidue
    int k = 0;
    while
          (k < this->size() && (*this)[k] == 0) k++; // 找前導零
    if (k == this->size()) return Poly(m); // 全零多項式
    if (k % 2 != 0) return Poly(); // 無解: 最低次項為奇數
    int s = quadraticResidue((*this)[k]);
    if (s == -1) return Poly(); // 無解: 係數無平方根
    int oft = k / 2, r = m - oft;
    if (r <= 0) return Poly(m);
    Poly h = this->shift(-k) * (*this)[k].inv(), g{1};
    for (int i = 1; i < r; i <<= 1) {
      int len = i << 1;
      g = (g + h.trunc(len) * g.inv(len)).trunc(len) / 2;
    }
    g.resize(r);
    g = (g * Z(s)).shift(oft).trunc(m);
```

```
    return g;
  }
  Poly exp(int m) const {
    Poly x{1};
    int k = 1;
    while (k < m) {
      k *= 2;
      x = (x * (Poly{1} - x.log(k) + trunc(k))).trunc(k);
    }
    return x.trunc(m);
  }
  Poly mulT(Poly b) const {
    if (b.empty()) return Poly();
    int n = b.size();
    reverse(b.begin(), b.end());
    return ((*this) * b).shift(-(n - 1));
  }
  vector<Z> eval(vector<Z> x) const {
    if (this->size() == 0) return vector<Z>(x.size(), 0);
    const int n = max(x.size(), this->size());
    vector<Poly> q(4 * n);
    vector<Z> ans(x.size());
    x.resize(n);
    function<void
        (int, int, int)> build = [&](int p, int l, int r) {
      if (r - l == 1) {
        q[p] = Poly{1, -x[l]};
      } else {
        int m = (l + r) / 2;
        build(2 * p, l, m);
        build(2 * p + 1, m, r);
        q[p] = q[2 * p] * q[2 * p + 1];
      }
    };
    build(1, 0, n);
    function<void(int, int, int, const Poly
        &)> work = [&](int p, int l, int r, const Poly &num) {
      if (r - l == 1) {
        if (l < int(ans.size())) ans[l] = num[0];
      } else {
        int m = (l + r) / 2;
        work(2
            * p, l, m, num.mulT(q[2 * p + 1]).resize(m - l));
        work(2
            * p + 1, m, r, num.mulT(q[2 * p]).resize(r - m));
      }
    };
    work(1, 0, n, mulT(q[1].inv(n)));
    return ans;
  }
};
template<int P, int G> vector<int> Poly<P, G>::rev;
template<int P, int G> vector<Mint<P>> Poly<P, G>::w = {0, 1};
```

10.3 FWT [73eea9]

```
void fwt(vector<ll> &a, bool inv = false) {
  // Or : [[1, 0], [1, 1]] -> [[1, 0], [-1, 1]]
  // And: [[1, 1], [0, 1]] -> [[1, -1], [0, 1]]
  // Xor: [[1, 1], [1, -1]] -> [[0.5, 0.5], [0.5, -0.5]]
  int n = __lg(int(a.size()));
  for (int i = 0; i < n; ++i) {
    for (int j = 0; j < 1 << n; ++j) {
      if (j >> i & 1) { // And
        ll x = a[j ^ (1 << i)];
        ll y = a[j];
        if (!inv) {
          a[j ^ (1 << i)] = x + y;
          a[j] = y;
        } else {
          a[j ^ (1 << i)] = x - y;
          a[j] = y;
        }
      }
    }
  }
}
```

10.4 Barlekamp Massey [853989]

```
// find s[i] = c[0]
      * s[i - 1] + c[1] * s[i - 2] + ... + c[k - 1] * s[i - k]
// return : c(x) = 1 - c_0x - c_1x^2 - ... - c_{k-1}x^k
template<int P = 998244353>
Poly<P> berlekampMassey(const Poly<P> &s) { // O(n^2)
  Poly<P> c, oldC;
  int f = -1;
  for (int i = 0; i < s.size(); i++) {
    auto delta = s[i];
    for (int j = 1; j <= c.size(); j++)
      delta -= c[j - 1] * s[i - j];
    if (delta == 0) continue;
    if (f == -1) {
      c.resize(i + 1);
      f = i;
    } else {
      auto d = oldC;
      d *= -1;
      d.insert(d.begin(), 1);
      Mint<P> df1 = 0;
      for (int j = 1; j <= d.size(); j++)
        df1 += d[j - 1] * s[f + 1 - j];
      assert(df1 != 0);
      auto coef = delta / df1;
      d *= coef;
      Poly<P> zeros(i - f - 1);
      zeros.insert(zeros.end(), d.begin(), d.end());
      d = zeros;
      auto temp = c;
      c += d;
      if (i - temp.size() > f - oldC.size()) {
        oldC = temp;
        f = i;
      }
    }
  }
  c *= -1;
  c.insert(c.begin(), 1);
  return c;
}
```

10.5 Linear Recurrence [95d930]

```
// a_i = c[0] *
      a_{i - 1} + c[1] * a_{i - 2} + ... + c[k - 1] * a_{i - k}
// q(x) = 1 - c[0] * x - c[1] * x^2 - ... - c[k - 1] * x^k
// p(x) = (a * q).trunc(c.size())
// return : a_n
template<int P = 998244353> // O(m log m log n)
Mint<P> linearRecurrence(Poly<P> p, Poly<P> q, ll n) {
  int m = q.size() - 1;
  while (n > 0) {
    auto nq = q;
    for (int i = 1; i <= m; i += 2) nq[i] *= -1;
    auto np = p * nq;
    nq = q * nq;
    for (int i = 0; i < m; i++) p[i] = np[i * 2 + n % 2];
    for (int i = 0; i <= m; i++) q[i] = nq[i * 2];
    n /= 2;
  }
  return p[0] / q[0];
}
```

11 Else

11.1 Segment Manipulation [635864]

```
auto era = [&](set<pair<int, int>> &st, int L, int R) -> void {
  auto it = st.lower_bound({L, -1});
  if (it != st.begin() && prev(it)->second >= L) --it;
  while (it != st.end() && it->first <= R) {
    auto [l, r] = *it;
    it = st.erase(it);
    if (l < L) st.emplace(l, L - 1);
    if (R < r) st.emplace(R + 1, r);
  }
};
auto ins = [&](set<pair<int, int>> &st, int L, int R) -> void {
  auto it = st.lower_bound({L, -1});
  if (it != st.begin() && prev(it)->second + 1 >= L) --it;
  while (it != st.end() && it->first <= R + 1) {
    L = min(L, it->first);
    R = max(R, it->second);
    it = st.erase(it);
  }
  st.emplace(L, R);
};
```

11.2 Python [7c66a4]

```
from decimal import * # 高精度浮點數
from fractions import * # 分數
from random import *
from math import *
# set decimal prec bigger if it could overflow in precision
setcontext
    (Context(prec=10, Emax=MAX_EMAX, rounding=ROUND_FLOOR))
# read and print
x = int(input())
a, b, c = list(map(Fraction, input().split()))
arr = list(map(Decimal, input().split()))
print(*arr)
# set
st = set(); st.add((a, b)); st.remove((a, b))
if not (a, b) in st:
# dict
d = dict(); d[(a, b)] = 1; del d[(a, b)]
for (a, b) in d.items():
# random
arr = [randint(l, r) for i in range(size)]
choice([8, 6, 4, 1]) # random pick one
shuffle(arr)
```

11.3 Fraction [62f33d]

```
template<class T> struct Fraction {
  T n, d;
  void reduce() {
    T g = gcd(abs(n), abs(d));
    n /= g, d /= g;
    if (d < 0) n = -n, d = -d;
  }
  Fraction(T n = 0, T d = 1) : n(n), d(d)
```

```
  { assert(d != 0); reduce(); }
  Fraction(const string &str) {
    char slash;
    if (str.find('/') != -1) {
      string x = str.substr(0, str.find('/'));
      string y = str.substr(str.find('/') + 1);
      n = stoBigint(x), d = stoBigint(y);
    } else {
      n = stoBigint(str), d = 1;
    }
    Fraction(n, d);
  }
  Fraction operator+(Fraction rhs) const
  { return Fraction(n * rhs.d + rhs.n * d, d * rhs.d); }
  Fraction operator-(Fraction rhs) const
  { return Fraction(n * rhs.d - rhs.n * d, d * rhs.d); }
  Fraction operator*(Fraction rhs) const
  { return Fraction(n * rhs.n, d * rhs.d); }
  Fraction operator/(Fraction rhs) const {
    assert(rhs.n != 0);
    return Fraction(n * rhs.d, d * rhs.n);
  }
  friend istream &operator>>(istream &is, Fraction &f) {
    string s; is >> s;
    f = Fraction(s);
    return is;
  }
  friend ostream &operator<<(ostream &os, const Fraction &f) {
    if (f.d == 1) os << f.n;
    else os << f.n << "/" << f.d;
    return os;
  }
  bool operator
       ==(Fraction b) const { return n * b.d == b.n * d; }
  bool operator
       !=(Fraction b) const { return n * b.d != b.n * d; }
  bool operator
       <(Fraction b) const { return n * b.d < b.n * d; }
};
```

11.4 Bigint [c581e1]

```
struct Bigint { // not support hex division
private:
  static const int digit = 9;  // hex: 7
  static const int base = 10;  // hex: 16
  static const int B = power(ll(base), digit);
  Bigint(vector<int> x, int sgn) : x(x), sgn(sgn) {}
  template<class U> vector<int> norm(vector<U> a) {
    if (a.empty()) return {0};
    for (int i = 0; i < a.size(); i++) {
      U c = a[i];
      a[i] = c % B;
      c /= B;
      if (c) {
        if (i == a.size() - 1) a.push_back(c);
        else a[i + 1] += c;
      }
    }
    while (a.size() > 1 && a.back() == 0) a.pop_back();
    return {a.begin(), a.end()};
  }
  void resign() { sgn = x.back() == 0 ? 1 : sgn; }
  int toInt(char c) const {
    if (isdigit(c)) return c - '0';
    else return c - 'A' + 10;
  }
  char toChar(int c) const {
    if (c < 10) return c + '0';
    else return c - 10 + 'A';
  }
public:
  int sgn = 1;
  vector<int> x; // 反著存
  Bigint() : x {0}, sgn(1) {}
  Bigint(ll a) { *this = Bigint(std::to_string(a)); }
  Bigint(string s) {
    if (s.empty()) { *this = Bigint(); return; }
    if (s[0] == '-') s.erase(s.begin()), sgn = -1;
    int add = 0, cnt = 0, b = 1;
    while (s.size()) {
      if (cnt == digit) {
        x.push_back(add), add = cnt = 0;
        b = 1;
      }
      add += toInt(s.back()) * b;
      cnt++, b *= base;
      s.pop_back();
    }
    if (add) x.push_back(add);
    x = norm(x);
  }
  int size() const { return x.size(); }
  Bigint abs() const { return Bigint(x, 1); }
  string to_string() const {
    string res;
    for (int i = 0; i < x.size(); i++) {
      string add; int v = x[i];
      for (int j = 0; j < digit; j++)
        add += toChar(v % base), v /= base;
      res += add;
    }
    while (res.size() > 1 && res.back() == '0') res.pop_back();
    if (sgn == -1) res += '-';
    reverse(res.begin(), res.end());
    return res;
  }
  Bigint operator-() const { return Bigint(x, -sgn); }
  friend Bigint operator+(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a - (-b);
    int n = max(a.size(), b.size());
    a.x.resize(n), b.x.resize(n);
    for (int i = 0; i < n; i++) a.x[i] += b.x[i];
    a.x = a.norm(a.x), a.resign();
    return a;
  }
  friend Bigint operator-(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a + (-b);
    if (a.abs() < b.abs()) return -(b - a);
    int n = max(a.size(), b.size());
    a.x.resize(n), b.x.resize(n);
    for (int i = 0; i < n; i++) {
      a.x[i] -= b.x[i];
      if (a.x[i] < 0) a.x[i] += B, a.x[i + 1]--;
    }
    a.x = a.norm(a.x), a.resign();
    return a;
  }
  friend istream &operator>>(istream &is, Bigint &a)
  { string v; is >> v; a = Bigint(v); return is; }
  friend ostream &operator<<(ostream &os, Bigint a)
  { os << a.to_string(); return os; }
  friend bool operator<(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a.sgn < b.sgn;
    if (a.x.size() != b.x.size()) {
      return a.x.size() < b.x.size();
    } else {
      for (int i = a.x.size() - 1; i >= 0; i--)
        if (a.x[i] != b.x[i]) return a.x[i] < b.x[i];
    }
    return 0;
  }
  friend bool operator>(Bigint a, Bigint b) {
    if (a.sgn != b.sgn) return a.sgn > b.sgn;
    if (a.x.size() != b.x.size()) {
      return a.x.size() > b.x.size();
    } else {
      for (int i = a.x.size() - 1; i >= 0; i--)
        if (a.x[i] != b.x[i]) return a.x[i] > b.x[i];
    }
    return 0;
  }
  friend bool operator==(const Bigint &a, const Bigint &b)
  { return a.sgn == b.sgn && a.x == b.x; }
  friend bool operator!=(const Bigint &a, const Bigint &b)
  { return a.sgn != b.sgn || a.x != b.x; }
  friend bool operator>=(const Bigint &a, const Bigint &b)
  { return a == b || a > b; }
  friend bool operator<=(const Bigint &a, const Bigint &b)
  { return a == b || a < b; }
};
Bigint abs(const Bigint &a) { return a.abs(); }
Bigint stoBigint(const string &s) { return Bigint(s); }
```

11.5 Multiple [a8c792]

```
// Require:
// Mint, NTT ~constructor and * operator
using i128 = __int128_t;
const int P1 = 1045430273; // 2^20 * 997 + 1
const int P2 = 1051721729; // 2^20 * 1003 + 1
const int P3 = 1053818881; // 2^20 * 1007 + 1
using Poly1 = Poly<1045430273, 3>;
using Poly2 = Poly<1051721729, 6>;
using Poly3 = Poly<1053818881, 7>;
const i128 T1 = i128(P2) * P3;
const i128 T2 = i128(P1) * P3;
const i128 T3 = i128(P1) * P2;
const int I1 = Mint<P1>(T1).inv().x;
const int I2 = Mint<P2>(T2).inv().x;
const int I3 = Mint<P3>(T3).inv().x;
const i128 M = i128(P1) * P2 * P3;
vector<i128> arbitraryMult
    (const vector<int> &a, const vector<int> &b) {
  int n = a.size(), m = b.size();
  Poly1 x
       = Poly1(a.begin(), a.end()) * Poly1(b.begin(), b.end());
  Poly2 y
       = Poly2(a.begin(), a.end()) * Poly2(b.begin(), b.end());
  Poly3 z
       = Poly3(a.begin(), a.end()) * Poly3(b.begin(), b.end());
  vector<i128> res(x.size());
  for (int i = 0; i < x.size(); i++)
    res[i] = (x[i].x * T1 % M * I1 +
          y[i].x * T2 % M * I2 +
          z[i].x * T3 % M * I3) % M;
  return res;
}
public:
  friend Bigint operator*(Bigint a, Bigint b) {
    a.x = a.norm(arbitraryMult(a.x, b.x));
    a.sgn *= b.sgn, a.resign();
    return a;
  }
```

11.6 Division [79e100]

```
private:
  vector<int> smallDiv(vector<int> a, int v) {
    ll add = 0;
    for (int i = a.size() - 1; i >= 0; i--) {
      add = add * B + a[i];
      int q = add / v;
      a[i] = q, add %= v;
    }
    return norm(a);
  }
  friend Bigint operator<<(Bigint a, int k) {
    if (!a.x.empty()) {
      vector<int> add(k, 0);
      a.x.insert(a.x.begin(), add.begin(), add.end());
    }
    return a;
  }
  friend Bigint operator>>(Bigint a, int k) {
    a.x = vector<int
         >(a.x.begin() + min(k, int(a.x.size())), a.x.end());
    a.x = a.norm(a.x);
    return a;
  }
public:
  friend Bigint operator/(Bigint a, Bigint b) {
    a = a.abs(), b = b.abs();
    a.sgn *= b.sgn;
    if (a < b) return Bigint();
    if (b.size() == 1) {
      a.x = a.smallDiv(a.x, b.x[0]);
    } else {
      Bigint inv = 1LL * B * B / b.x.back();
      Bigint pre = 0, res = 0;
      int d = a.size() + 1 - b.size();
      int cur = 2, bcur = 1;
      while (inv != pre || bcur < b.size()) {
        bcur = min(bcur << 1, b.size());
        res.x = {b.x.end() - bcur, b.x.end()};
        pre = inv;
        inv = inv
            * ((Bigint(2) << (cur + bcur - 1)) - inv * res);
        cur = min(cur << 1, d);
        inv.x = {inv.x.end() - cur, inv.x.end()};
      }
      inv.x = {inv.x.end() - d, inv.x.end()};
      res = (a * inv) >> a.size();
      Bigint mul = res * b;
      while (mul + b <= a) res = res + 1, mul = mul + b;
      a.x = a.norm(res.x);
    }
    return a;
  }
  friend Bigint operator%(Bigint a, Bigint b)
  { return a = a - (a / b) * b; }
Bigint gcd(Bigint a, Bigint b) {
  while (b != 0) {
    Bigint r = a % b;
    a = b, b = r;
  } return a;
}
```

11.7 Division-Python [110bd8]

```
from decimal import * # 無誤差浮點數
setcontext(
     Context(prec=4000000, Emax=4000000, rounding=ROUND_FLOOR))
t = int(input())
for i in range(t):
  a, b = map(Decimal, input().split())
  d, m = divmod(a, b)
  print(d, m)
```